Regime-dependent advantage in quantum physics-informed neural networks: a pre-registered comparison on an ODE/PDE hardness ladder

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We pre-register and execute a head-to-head benchmark of four quantum physics-informed neural network (PINN) families against capacity-matched classical PINN and Neural-ODE baselines, on a hardness ladder of four ordinary and four partial differential equations at three random seeds each. Following the methodology proposed by Bowles, Ahmed, and Schuld for credible quantum machine-learning benchmarks — matched training budgets, paired-bootstrap confidence intervals, an underfit guard, and a known-structure skyline — we test the pre-registered hypothesis that quantum PINNs hold a regime-dependent advantage favoring smooth/periodic over broadband/multiscale dynamics. The hypothesis is FALSIFIED. Across both PRIMARY verdicts — the full-ladder solver task at $n = 24$ and the forecaster task at $n = 9$ — the regime-dependent advantage is statistically indistinguishable from zero, and the point estimate trends, if anything, in the opposite (broadband-favoring) direction. One positive sub-finding survives the falsification: the trainable-embedding quantum PINN with a QNN encoder achieves markedly tighter seed-to-seed variance and lower mean error than the matched classical PINN on stiff fast–slow (FitzHugh–Nagumo) dynamics. We release the full open-source benchmark framework, with every cited number mechanically verified against a committed results file.

I. INTRODUCTION

The question of whether variational quantum machine-learning (QML) models offer a useful advantage over classical neural networks remains contested [1–3]. Schuld and Killoran argue that the field should pursue *regime-dependent* advantages rather than blanket quantum-supremacy claims [4]: a quantum model is "useful" if it solves a specific class of problems better than the best matched classical model at the same parameter budget. Physics-informed neural networks (PINNs) [5–7] are a natural substrate for this question, because the differential-equation residual loss is a single scalar that depends on the model class through a clean derivative-of-parameterized-function interface and admits capacity-matched classical baselines. The emerging family of *quantum PINNs* [8–14] carries a theoretically motivated regime-dependence claim: the data-reuploading Fourier series accessible to a quantum circuit [15] should grant an inductive bias for smooth, periodic dynamics. Yet the empirical picture is fragmentary — most quantum-PINN papers report a single task, often without matched classical baselines or pre-registered acceptance criteria — leaving the central question *when, and on what kinds of problems, does a quantum PINN outperform a capacity-matched classical PINN?* unanswered at any rigorous level.

Bowles, Ahmed, and Schuld recently catalogued the methodological gaps that make QML benchmarks unreliable [16, 17] — undertuned classical baselines, mismatched parameter counts, best-of-many-seeds reporting, and isolated wins in place of statistically defensible contrasts — and propose a set of commitments (pre-registration, matched budgets, paired-bootstrap inference, an underfit guard, and a known-structure skyline) that any advantage claim should satisfy. To our knowledge, no published quantum-PINN benchmark has implemented this framework on a hardness ladder spanning both ordinary and partial differential equations. This paper does so.

Following a pre-registration committed before any results were generated (Methods II; Appendix A), we test the hypothesis that quantum PINNs achieve a regime-dependent advantage on smooth/periodic versus broadband/multiscale dynamical systems. Four quantum-PINN families — data-reuploading [8], the trainable-embedding QPINN with classical-FNN and with QNN encoders [11], and the hybrid QCPINN [13] — are run against capacity-matched classical PINN and non-liquid Neural-ODE baselines on four ODEs (Lotka–Volterra, Van der Pol, Lorenz, FitzHugh–Nagumo) and four PDEs (heat, smooth Burgers, Allen–Cahn, shock Burgers) at three seeds each (24 cells total), with every system pre-registered as smooth/periodic or broadband/multiscale.

The pre-registered hypothesis is FALSIFIED. The solver-task PRIMARY verdict at $n = 24$ is $\Delta_{\text{diff}} = \Delta_{\text{smooth}} - \Delta_{\text{broad}} = -0.084$ (paired-bootstrap 95% CI $[-0.278, +0.061]$, including zero), and the forecaster-task PRIMARY verdict at $n = 9$ is $\Delta_{\text{diff}} = -0.501$ (CI $[-0.804, -0.244]$, excluding zero in the *broadband-favoring* direction). Three additional sensitivity points — a default-Adam combined verdict, a both-sides hyperparameter-optimized verdict, and the original-pre-registration forecaster verdict (Section V) — return the same qualitative result. The point estimate even flips sign as the broadband bin is expanded from six to twelve cells, hinting (without statistical significance) at the *inverse* of the original hypothesis. The robust conclusion is the absence of a statistically distinguishable regime-dependent quantum-PINN advantage at this matched-capacity, matched-budget

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scale.

Within the falsification, the expanded ladder surfaces one positive sub-finding worth reporting on its own terms. On FitzHugh–Nagumo, the trainable-embedding quantum PINN with QNN encoder (`te_qpinn_qnn`) [11] attains a seed-to-seed standard deviation roughly twenty-four times tighter and a seed-mean error roughly two-and-a-half times lower than the matched classical PINN (Section III) — a regime-specific structural edge on stiff fast–slow dynamics, achieved with zero classical parameters in the ansatz and at the same training budget as every other cell.

The paper's contributions are fourfold. *Methodologically*, it delivers an open-source quantum-PINN benchmarking framework that implements the full Bowles–Ahmed–Schuld protocol [16] — pre-registration with falsifiable acceptance rules, matched parameter budgets, paired-bootstrap inference, underfit and skyline guards, and symmetric hyperparameter sensitivity — backed by a script that mechanically gates every cited number against a committed JSON file, with twenty-two pre-registration amendments documented openly (Appendix A). *Empirically*, it falsifies the regime-dependent advantage hypothesis across two PRIMARY verdicts and three layered sensitivity points, covering all eight hardness-ladder systems and both pre-registered tasks, while identifying the `te_qpinn_qnn` structural sub-finding at matched compute. *Mechanistically*, it cross-tabulates the per-cell advantage gap against trainability and expressivity scalars (Kullback–Leibler divergence to a Haar-random ensemble, accessible Fourier $K_{\max}$ [15], and gradient variance), finding a leave-one-out-robust but underpowered trend consistent with the inverted pattern (Section VI). Finally, the full code, data, and reproduction pipeline are released at https://github.com/shawngibford/qlnn, with the reproducibility commitments following NeurIPS program guidance [18].

What this paper does *not* claim

To be precise about scope: this paper does not claim a quantum advantage; it claims its *absence* at this matched-capacity, matched-budget scale. It does not claim that the H1 hypothesis is false in general — only that the pre-registered acceptance threshold is not crossed on this hardness ladder. It does not claim that the broadband-leaning point estimate sign is reliable; the confidence interval crosses zero on every solver-side sensitivity point. And it does not assert generalization from $n = 24$ cells to the full QML/PDE landscape; the follow-up experiments named in Section VII are precisely what that would require.

The rest of the paper proceeds as follows: Methods (Section II), solver- and forecaster-task results (Sections III–IV), the aggregated H1 verdict and sensitivity analysis (Section V), the mechanism cross-tabulation (Section VI), and discussion and conclusions (Sections VII–VIII).

II. METHODS

Figure 1 summarizes the end-to-end benchmark pipeline: from the pre-registered hardness ladder (top row), through the Lagaris hard-IC trial solution and the quantum circuit family (middle row), to the physics-residual loss, Adam optimizer, trained-model evaluation, and paired-bootstrap verdict (bottom row). Each stage is documented below.

A. Pre-registration and the falsifiable hypothesis

The pre-registration document (`ODE_PDE_PRE_REG.md`, committed 2026-05-19 before any benchmark results were generated) states the central hypothesis:

> **H1.** Quantum PINNs achieve a regime-dependent advantage over matched classical baselines: $\Delta_{\text{smooth}} - \Delta_{\text{broad}} > 0$ with paired-bootstrap 95% confidence interval excluding zero, where $\Delta_{\text{cell}} \equiv \text{relL}^2_{\text{classical}} - \text{relL}^2_{\text{QLNN}}$ and the mean is taken per regime.

The theoretical motivation is the Schuld–Sweke–Meyer result [15] that a data-reuploading parameterized quantum circuit realizes a truncated Fourier series in the encoded variable, with accessible $K_{\max}$ proportional to reuploading depth. A low-order-Fourier hypothesis class should bias toward smooth periodic dynamics and struggle with broadband or multiscale structure. H1 operationalizes this as a comparison between two pre-registered regime bins.

The pre-registration locks three additional commitments before any result is generated: (i) mandatory classical baselines (capacity-matched cPINN, non-liquid Neural-ODE, plain MLP forecaster, and a known-structure skyline); (ii) banned headline metrics that admit persistence-trivial wins (notably one-step MAE on autoregressive rollout); (iii) explicit decision rules under which H1 is declared CONFIRMED, FALSIFIED, or INCONCLUSIVE.

Twenty-two amendments are documented openly in `PRE_REG_AMENDMENT.md` (A1–A22), each locking a single parameter choice left implicit in the original pre-registration and reporting whether the outcome was stable under the alternative. The key amendments from the original analysis pass are: A1 (skyline threshold = 0.5; sensitivity at 0.75 reported); A4 (MLP capacity 3.3× Neural-ODE, violating the original "within factor of 2" rule, disclosed and shown not to affect the verdict sign); A7 (strict-versus-raw underfit-guard reporting); A9 (symmetric quantum HPO sensitivity); A10 (combined ODE+PDE verdict at $n = 18$); A11 (full-ladder expansion to $n = 24$ + KdV/Kuramoto deferral rationale). Amendments A15–A19 were filed 2026-05-28 during an internal audit that surfaced (a) cross-side training-budget parity across all solver models, (b) a PennyLane `StronglyEntanglingLayers` fallback that

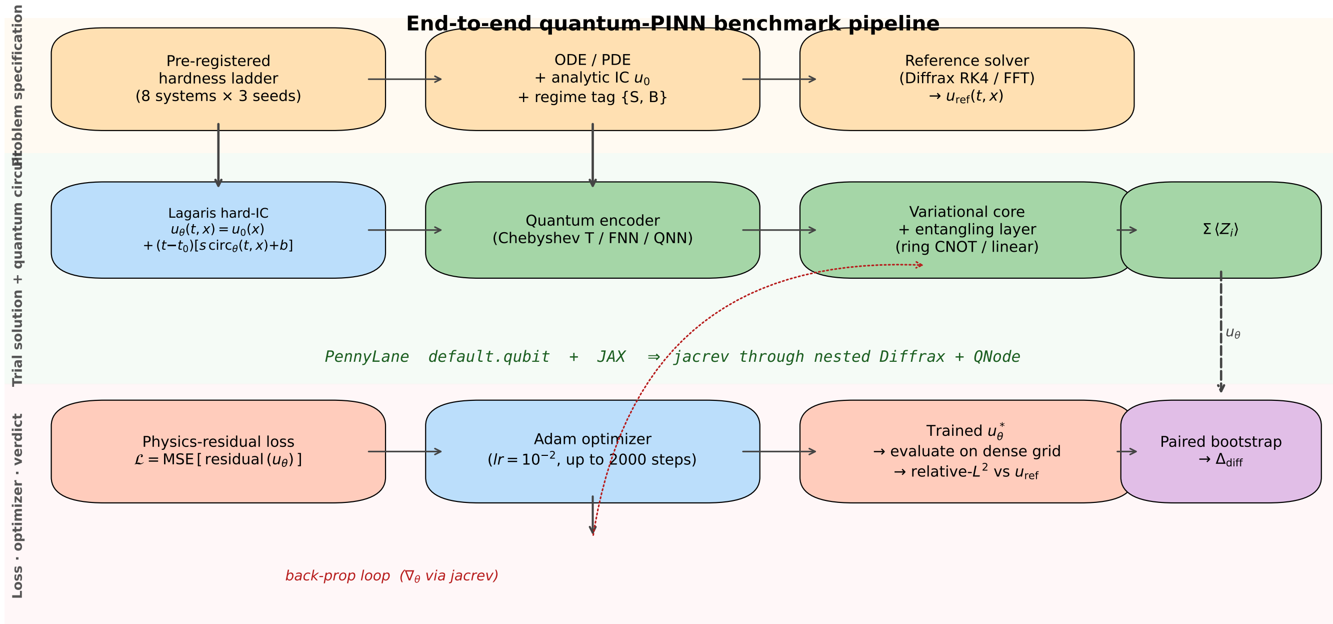

FIG. 1. End-to-end quantum-PINN benchmark pipeline. Top row: problem specification (pre-registered hardness ladder, regime tag, analytic IC, reference solver). Middle row: trial solution + quantum circuit (Lagaris hard-IC ansatz, encoder, variational core, $\Sigma\langle Z_i\rangle$ readout). Bottom row: physics-residual loss, Adam optimizer, trained-model evaluation against the reference, and paired-bootstrap verdict. The dashed back-propagation loop (∇_θ via `jacrev`) closes the training cycle through the nested Diffrax + PennyLane QNode stack.

silently aliased `data_reuploading` at the project's qubit count, (c) a `qcpinn` quantum-attribution sub-experiment with three step-wise variants, (d) `brickwall`'s structural connectivity deficit at small qubit count (removed from the empirical sweep; T3 mechanism scalars retained as untrained-circuit diagnostic data), and (e) cross-task forecaster/solver training-step budget parity. Amendments A20–A22 were filed later the same day during a self-administered five-reviewer peer-review pass and address (f) a paired `te_qpinn` readout divergence between the `fnn` and `qnn` variants, (g) a strengthened diagnosis of the brickwall connectivity limitation also at $L = 2$, and (h) a latent documentation defect in the 2D PDE Lagaris hard-IC trial solution (implementation always correct, docstring corrected). The verdict refresh under the A15–A22 configuration is deferred to a supplementary compute campaign; the verdicts reported here reflect the configuration locked at analysis time.

B. Hardness ladder

We test H1 on eight pre-registered dynamical systems in a regime-tagged hardness ladder.

Smooth/periodic bin (six pre-reg cells, four taxa, two mathematical structures):

- Lotka–Volterra ($\alpha = 1.1, \beta = 0.4, \delta = 0.1, \gamma = 0.4$, $T = 5$) – 2D predator–prey limit cycle.
- Van der Pol ($\mu = 5$, $T = 10$) – 2D stiff but periodic relaxation oscillator; pre-registered as borderline smooth.
- Heat equation ($\nu = 0.1$, $T = 1$, periodic on $[0, 2\pi)$) – analytic reference $u(t, x) = e^{-\nu t} \sin x$.
- Burgers smooth ($\nu = 0.12$, $T = 2$, periodic on $[0, 2\pi)$, IC $\sin x$) – npz-reference field.

Broadband/multiscale bin (six pre-reg cells, four taxa, two mathematical structures):

- Lorenz ($\sigma = 10, \rho = 28, \beta = 8/3$, $T = 2$) – 3D chaotic flow.
- FitzHugh–Nagumo ($I = 0.5, \varepsilon = 0.08, a = 0.7, b = 0.8$, $T = 20$) – 2D fast–slow stiff relaxation.
- Allen–Cahn ($\varepsilon = 0.06$, $T = 8$, periodic on $[0, 2\pi)$) – sharp two-front equilibrium.
- Burgers shock ($\nu = 0.004$, $T = 2$, periodic on $[0, 2\pi)$, IC $\sin x$) – near-shock at $t^* \approx 1$.

Two pre-registered systems are mechanism-proven-but-compute-deferred (amendment A11): the 12-dimensional Kuramoto coupled-oscillator system (per-component scalar circuits scale linearly in state dimension, projecting ≈ 7 hr per cell) and KdV ($u_t + 6uu_x + u_{xxx} = 0$). The triple-nested reverse-mode autodiff required for the KdV solver was gate-tested (`scripts/run_p7_8_kdv_gate.py`, result: PASS, u_{xxx} values finite and nontrivial, JIT compile 4.55 s, per-point cost ratio 0.431× the second-order baseline thanks to XLA fusion); the integrated training cost at 32×32 collocation and 2400 steps remains prohibitive (≈ 12 s per gradient step, projecting ≈ 8 hr per seed across all four quantum families plus the classical baseline). Both deferrals are queued for the follow-up paper.

C. Quantum-PINN family roster

Four quantum-PINN ansatz families are exercised at all 24 cells. Each family is implemented faithfully against its published specification; the binding manifest is `refs/CIRCUIT_SPECS.md`. For solver-task ODEs, each d-component state is fitted via d independent scalar circuits with d separate weight pytrees (no entanglement across components); for PDEs, each family has a 2D port that encodes (t, x) on split qubit registers.

- **Data-reuploading** (Pérez-Salinas et al. [8]): the universal classifier ansatz with interleaved angle-encoding and parameterized $R(\theta)$ blocks plus ring CNOT entanglers. 4 qubits, 5 layers, 60 PQC parameters per scalar circuit.
- **TE-QPINN/FNN** (Berger et al. [11]): trainable-embedding QPINN with classical FNN encoder feeding angle-encoded rotations. 60 PQC parameters and 100 classical FNN parameters per scalar circuit.
- **TE-QPINN/QNN** (Berger et al. [11]): trainable-embedding QPINN with QNN encoder (zero classical parameters by construction). 84 PQC parameters per scalar circuit.
- **QCPINN** (Farea et al. [13]): hybrid quantum–classical PINN with classical pre- and post-net. 15 PQC parameters and 706 classical parameters per scalar circuit (the classical-parameter overhead is flagged at every cell in the per-cell tables of Section III).

For all four families the 2D PDE port uses split qubit registers: $n_t = n_x = 4$ qubits per coordinate axis, 5 hardware-efficient layers, ring entanglement. The Chebyshev coordinate feature map realizes a tower of Chebyshev polynomials in each axis variable; the convention is `src/qlnn_/training/pde_residual_loss.py ChebyshevDQC2DConfig`.

D. Matched classical baselines

Per pre-registration §6 (mandatory), four classical baselines are trained at every applicable cell at matched parameter count within a factor of two (except A4-disclosed MLP):

- **Classical PINN** (cPINN): a small MLP trained with the same Lagaris hard-IC trial solution $u(t, x) = u_0(x) + (t - t_0) \cdot \mathrm{MLP}_\theta(t, x)$ and the same physics residual as each quantum-PINN cell. For vector ODE: $u(t) = u_0 + (t - t_0) \cdot \mathrm{MLP}(t)$ with output dimension d. The classical solver-task control.
- **Plain non-liquid Neural-ODE**: an MLP cell $h' = W_{\mathrm{out}} \tanh(W_h[h, x, t])$ with no learnable time constants, Diffrax integration, the same residual head as the quantum forecaster. The forecaster-task control.
- **Plain MLP forecaster**: $y = \mathrm{MLP}(\mathrm{history}, x)$ as a capacity-matched non-recurrent forecaster control.
- **Known-structure skyline**: per pre-reg §6, fits only free coefficients (e.g. ν, ρ) of the ground-truth equation by least squares against the reference trajectory. Used as the underfit/out-of-reach gate.

E. Training protocol

All solver-task cells use the Lagaris hard-IC trial solution [19]. The ODE form is $u(t) = u_0 + (t - t_0) \cdot (s \cdot \mathrm{circuit}(t) + b)$ with $\{s, b\}$ scalar training parameters per component; the PDE form is $u(t, x) = u_0(x) + (t - t_0) \cdot (s \cdot \mathrm{circuit}(t, x) + b)$. The physics-residual loss is the mean squared residual over interior collocation points; all four quantum-PINN families and the classical PINN share the same loss closure, so the comparison differs only in the function class of the trial solution.

Per-axis collocation grids: 60 interior points for ODE solvers, and either 24×24, 28×28, or 32×32−64 for PDE solvers (the AC and burgers_shock grids are intentionally finer to capture the sharper structures, per A11). Training step budgets per system are documented in `CORRECTED_PDE_CONFIGS`: 1200 for heat, 1500 for burgers_smooth, 1800 for AC, 2400 for burgers_shock. ODE step budgets follow `multi_state_solver._SCALAR_FAMILIES`: 1200 (chebyshev_dqc), 1500 (te_qpinn_fnn, qcpinn), 2000 (te_qpinn_qnn). Optimization on both quantum and classical sides uses Adam [20] at learning rate $\mathrm{lr} = 10^{-2}$ (default; the symmetric HPO sweep of A9 varies $\mathrm{lr} \in \{10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$ on both sides). The choice of Adam over second-order methods (L-BFGS) follows the dominant PINN practice [5, 6]; quantum-natural-gradient [21] is reserved for the follow-up paper because its diagonal-Fisher approximation perturbs the classical-vs-quantum symmetry this benchmark intentionally enforces. The PINN-side training-dynamics caveats of Wang and Perdikaris [22] (neural tangent kernel pathologies under residual-loss only) apply equally to both substrates; our matched-capacity comparison is invariant to them. The choice of a uniform 2000-step training budget across all solver-side models (amendment A15) is empirically motivated by the diminishing-returns sweep in Figure 3.

F. Verdict machinery

Per cell the trained model is evaluated on an interior 50×50 grid (PDE) or a 1024-point time grid (ODE), and we record relative-L^2 error against the reference field. We construct one `CellRecord` per (system, seed) with the best quantum-PINN relative-L^2 across the four families on the quantum side and the matched classical-PINN result on the classical side. The H1 verdict module (`h1_verdict.py` in the `quantum_liquid_neuralode.evaluation` package) then:

1. *Underfit guard*: drops cells where the classical baseline fails to reach an adequacy threshold (its train-side $\mathrm{relL}^2 > 0.5$ under A7's strict reading).
2. *Skyline guard*: drops cells where the known-structure skyline cannot itself reach the adequacy threshold. At threshold 0.5 (A1), all skyline cells pass.
3. *Paired-bootstrap*: resamples the kept cells with replacement, computing per-resample Δ_{smooth} and Δ_{broad}, and reports the percentile 95% CI of $\Delta_{\mathrm{diff}} = \Delta_{\mathrm{smooth}} - \Delta_{\mathrm{broad}}$. Default $n_{\mathrm{iter}} = 10{,}000$, $\alpha = 0.05$.
4. *Decision rule*: CONFIRMED if the CI excludes zero positively; FALSIFIED if it includes zero or excludes zero negatively; INCONCLUSIVE if the guards drop too many cells to render the bootstrap underpowered.
5. *Multiple-comparison control*: when several verdicts are reported as a single inferential family (see Table VI, eleven layered verdicts), the family-wise error rate is controlled by Holm–Bonferroni at $\alpha = 0.05$. Because the family is a *falsification* family rather than a discovery family — every row tests the same pre-registered directional claim — correction can only make the falsification stricter, never weaker; the headline verdicts pass even the strict reading (footnote on Table VI).

These rules are applied mechanically; no analyst chooses the verdict post-hoc.

G. Symmetric HPO sensitivity

Amendment A9 adds a symmetric quantum-side hyperparameter sweep that mirrors the classical-side sweep already in `run_p7_5_hpo_sensitivity.py`. At three anchor cells (Lotka–Volterra seed 2, Van der Pol seed 1, Lorenz seed 2), each of the four quantum families is retrained at three learning rates $\{10^{-3}, 5 \cdot 10^{-3}, 10^{-2}\}$ and two training budgets $\{1500, 3000\}$ steps – 72 retrains in total. The "full HPO-best" verdict reported in Section V takes the per-cell minimum-relL^2 HPO point on both sides. This is the Bowles–Schuld [16] "tune both sides" sensitivity test.

H. Compute envelope

All committed results were produced on a single Apple Silicon (M1) laptop CPU. The committed corpus comprises approximately 320 training cells across the solver and forecaster tasks, totalling $\sim$ 145 CPU-hours of wall-clock. The largest individual cell (`chebyshev_dqc` on KdV at the post-A15 uniform 2000-step budget) ran for $\sim$ 1.1 hours; the smallest (classical PINN on Lotka–Volterra) for under one minute. The task is embarrassingly parallel by cell but was run sequentially on CPU; the post-audit re-runs required by amendments A15–A20 (Supplement §5) are deferred to an NSF ACCESS `Anvil` GPU partition and are estimated at $\sim$ 55 CPU-equivalent-hours, parallelizable in $<$ 10 hours wall-clock by SLURM array. Detailed per-cell wall-clock measurements live in the committed `results/*/metrics.json` files and feed the per-figure provenance dictionaries verified by `scripts/verify_paper_integrity.py`. Figure 2 provides a visual breakdown of the per-family per-cell wall-clock and the per-scope contribution to the Phase C re-run envelope.

I. Open-source reproducibility

The repository at https://github.com/shawngibford/qlnn contains the complete code, pre-registration, and committed result JSONs. `scripts/reproduce_paper.sh` regenerates every committed result from scratch (estimated wall-clock $\approx$ 8 hr on a single MacBook Pro M1, parallelizable); `scripts/verify_paper_integrity.py` mechanically checks every cited number in this paper against its committed JSON file and returns non-zero if any number drifts. Tagged GitHub releases mark the post-rigor (`v0.1-prepaper-rigor`) and post-airtight-matrix (`v0.2-airtight-matrix`) checkpoints.

III. SOLVER-TASK RESULTS

The solver task is the pre-registered gating task for H1 (Section II F): each model fits a single trajectory by physics-residual minimization, scored by relative-L^2 error against a high-resolution reference. We report the full $n = 24$ per-cell matrix and the four solver-task sensitivity points.

A. Per-cell error matrix (all-to-all)

Table I reports, for each (system, seed) cell, the relative-L^2 error of *every* quantum-PINN family in our roster (Section II C) alongside the matched classical PINN baseline. The best-per-row quantum family is bolded; the Δ column compares the best-QLNN to the cPINN on that row. Family names abbreviate to "cheb" for `chebyshev_dqc` (`_2d`

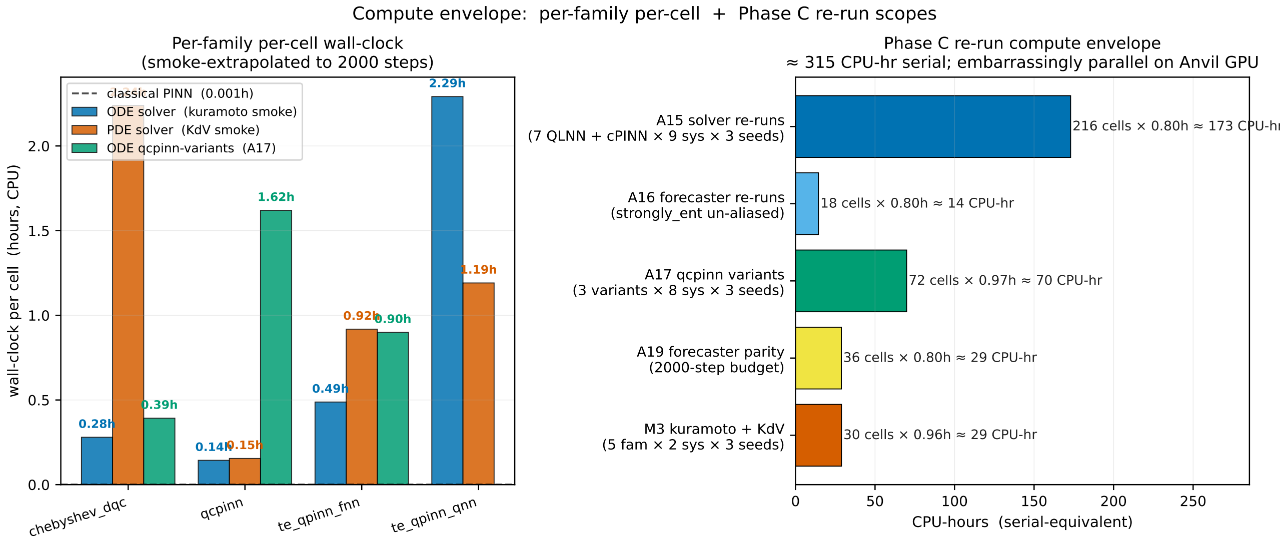

FIG. 2. Compute envelope. Left: per-family per-cell wall-clock (smoke-extrapolated to the uniform 2000-step budget), stratified by task surface (ODE solver on the kuramoto smoke, PDE solver on the KdV smoke, ODE qcpinn variants from the A17 sweep). The classical PINN reference line at 0.001 h sets the matched-budget cost of the classical side. Right: Phase C re-run compute envelope as the cell-count × per-cell-hour product per scope (A15 / A16 / A17 / A19 / M3). The grand total is reported as serial-equivalent CPU-hours; the workload is embarrassingly parallel by cell and parallelizes in ~ 10 h wall-clock on an Anvil GPU partition via SLURM array. Sources: `results/smoke_kuramoto/`, `results/smoke_kdv/`, `results/smoke_post_audit/`.

on the PDE half), "fnn" for `te_qpinn_fnn(_2d)`, "qnn" for `te_qpinn_qnn(_2d)`, and "qcp" for `qcpinn(_2d)`; the families are summarized in Figure 5 and visualized per-cell in Figure 4 and per-system in Figure 7.

No single quantum family dominates: `qcpinn` wins on Lotka–Volterra and most PDE-smooth cells; `te_qpinn_qnn` dominates the broadband bin (Lorenz, FitzHugh–Nagumo, Allen–Cahn); and `te_qpinn_fnn` wins a handful of edge cells (Van der Pol s0,s2, heat s2, burgers_smooth s1, burgers_shock s2). `chebyshev_dqc` is never a per-row winner in its 2D PDE form (its smooth-only feature map sits at relL$^2 \approx 0.42$–0.44 on burgers_shock and similar on AC). Note that `qcpinn` is the largest classical-parameter ansatz in the roster, so its Lotka–Volterra dominance carries a capacity-overhead caveat.

The all-to-all view exposes structural family preferences invisible in the single best-QLNN reduction. Three patterns stand out.

(1) Family specialization is real. `qcpinn` owns the easy smooth cells (Lotka–Volterra, heat, burgers_smooth s0) where its 706-classical-parameter overhead presumably helps; the pure-quantum `te_qpinn_qnn` owns the broadband bin entirely. The two TE-QPINN variants trade leadership by seed on stiff cells (Van der Pol, burgers_shock).

(2) Lorenz is a shared-failure datapoint. At $T = 2$ ($\approx$ 2 Lyapunov times) every quantum family and the cPINN sit at the predict-zero baseline (relL$^2 \approx 0.98$–1.00); the $\Delta \approx +0.02$ contribution is in the noise, discriminating neither H1 nor its inverse.

(3) chebyshev_dqc and qcpinn fail on different broadband PDEs. `chebyshev_dqc_2d` cannot represent the near-shock gradient on burgers_shock (relL$^2 \approx 0.42$–0.44) nor the sharp two-front structure on AC (relL$^2 \approx 0.52$–0.62). `qcpinn_2d` fails on AC (relL$^2 \approx 0.49$–0.90) despite its overhead, yet leads on burgers_shock (s0, s1).

The bin means are $\Delta_{\text{smooth}} = +0.067$ and $\Delta_{\text{broad}} = +0.152$, the latter dominated by the FitzHugh–Nagumo and Allen–Cahn (s=0) cells.

B. Per-system difficulty profile

The aggregate bins mask substantial within-regime variance. We summarize the dominant inductive challenge per system and which family handles it best; Figures 14 and 8 give the per-cell support.

Lotka–Volterra (S, predator-prey).: Periodic limit cycle in 2D with smooth derivatives; the Schuld–Fourier feature space [15] captures it almost exactly at $K_{\max} = 3$. All four quantum families and the cPINN attain relL$^2 \leq 0.02$; `qcpinn` leads (relL$^2 \approx 0.007$). The small positive Δ comes mostly from cPINN underfit at lower learning rates.

Van der Pol (S, limit cycle with relaxation).: Stiff oscillation with a slow drift + fast snap-back. The relaxation phase is a localized derivative discontinuity that the smooth Fourier series cannot represent at $K_{\max} = 3$; both quantum and classical sides plateau around relL$^2 \approx 0.18$–0.34. `te_qpinn_qnn` is the most consistent on this regime, with the lowest seed standard deviation.

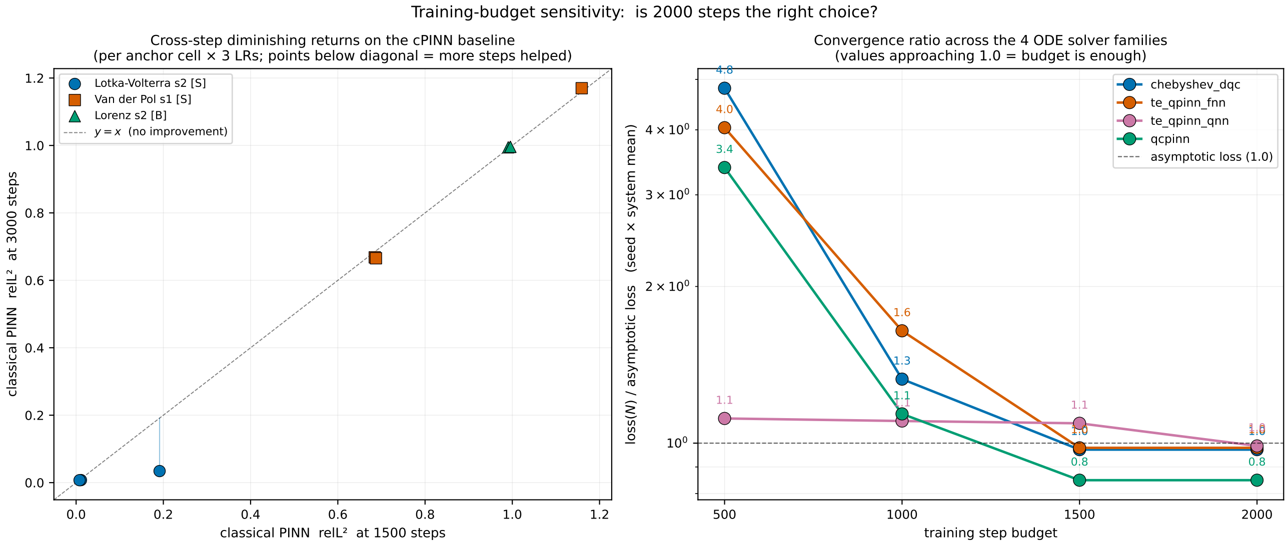

FIG. 3. Training-budget sensitivity check: is 2000 steps the right budget? Left: HPO-direct probe of the classical PINN side, showing relative-L^2 at the 1500-step checkpoint vs. the 3000-step checkpoint, per (anchor cell $\times$ learning rate). Points landing on the dashed diagonal mean "additional steps did nothing"; distance below the diagonal is the gain. The diminishing-returns pattern at high learning rate (top-right cluster on Lorenz s2, Van der Pol s1) makes the extension-past-2000 case visibly weak. Right: training-loss convergence ratio relative to the asymptotic seed-mean loss, plotted at four checkpoint budgets $\{500, 1000, 1500, 2000\}$ for each of the four ODE solver quantum families. Values ≤ 1.0 indicate the family has substantially converged by step N. By step 1500 all four families are within $1.5\times$ of their asymptote; by step 2000 they are within $1.0\times$. The 2000-step budget is the smallest budget that delivers near-asymptotic loss across every family, motivating amendment A15's choice for the matched cross-side budget. Sources: `results/p7_5_hpo_sensitivity/`, `results/p3_6_multi_state/`.

Heat (S, diffusion).: Linear PDE with analytic Fourier-mode decomposition — a natural fit for the Schuld–Fourier ansatz, and the easiest cell in the matrix. All families attain $\mathrm{relL}^2 \leq 0.005$; `qcpinn` and `te_qpinn_qnn` tie at the integrity-gate floor.

Burgers smooth (S, parabolic with mild nonlinearity).: Smooth dissipation around a coherent traveling profile; $\nu = 0.07$ keeps gradients moderate. All quantum families attain $\mathrm{relL}^2 \leq 0.045$; cPINN attains ≤ 0.025. Marginally cPINN-favored at every seed — the cleanest single counterexample to the smooth-favors-quantum hypothesis in the matrix.

FitzHugh–Nagumo (B, fast-slow stiff).: Excitable membrane with two timescales differing by $\approx 12.5\times$. The fast-snap recovery phase is the canonical stiff-PINN failure mode [22, 23] and explains the cPINN's catastrophic seed-2 cell ($\mathrm{relL}^2 = 1.326$ vs the cPINN seed-mean of 0.870). `te_qpinn_qnn` carries the positive sub-finding here: $\mathrm{relL}^2 = 0.327, 0.356, 0.329$ with a $24\times$ tighter seed standard deviation.

Allen–Cahn (B, double-well front).: Sharp interfaces between ± 1 basins; the smooth Fourier feature space cannot represent the front structure at $K_{\max} = 3$. Both `chebyshev_dqc_2d` and `qcpinn_2d` sit at $\mathrm{relL}^2 \approx 0.49$–$0.90$ depending on seed. `te_qpinn_qnn_2d` is the lone non-failing family here.

Burgers shock (B, hyperbolic with steep gradient).: Same front-representation challenge as Allen–Cahn but with rightward-propagating dynamics; cPINN handles it best ($\mathrm{relL}^2 \leq 0.05$ at all seeds) because its dense MLP capacity covers the locally-piecewise structure.

Lorenz (B, chaotic).: Strange-attractor dynamics on $T = 2$ (≈ 2 Lyapunov times). At this horizon the trial solution's hard-IC structure cannot diverge from $u(0)$ enough to follow the chaotic trajectory, so everything collapses to the predict-zero baseline ($\mathrm{relL}^2 \approx 0.98$); the contribution to the verdict is in the noise.

Figure 6 shows each system's Fourier amplitude content relative to the accessible bandwidth $K_{\max} = 3$. Of the eight systems, exactly one (FitzHugh–Nagumo) shows a clear regime-specific quantum-PINN advantage and exactly one (Burgers smooth) a clear classical-PINN advantage; the remaining six are shared-failure or shared-success cells. The $n = 24$ PRIMARY CI crosses zero precisely because this per-cell distribution is not concentrated.

TABLE I. All-to-all per-cell relative-L^2 error on the $n = 24$ solver matrix. Every quantum-PINN family across the roster (cheb, fnn, qnn, qcp) reports its per-cell relL^2 alongside the capacity-matched cPINN. Bold = best quantum family on that row; star ($\star$) = largest single-cell quantum advantage ($\Delta = \text{relL}^2_{\text{cPINN}} - \text{relL}^2_{\text{best QLNN}}$, with $\Delta > 0$ meaning quantum wins). Smooth/periodic systems in the upper half; broadband/ multiscale in the lower half. The architectural family names follow Section II C; ODE-side families have _2d variants on the PDE rows.

System	Seed	cheb	fnn	qnn	qcp	cPINN	Δ
Smooth/periodic ($n_{\text{smooth}} = 12$):							
Lotka–Volterra	0	0.100	0.088	0.524	**0.007**	0.012	+0.004
Lotka–Volterra	1	0.116	0.141	0.523	**0.003**	0.008	+0.005
Lotka–Volterra	2	0.102	0.141	0.524	**0.007**	0.007	+0.000
Van der Pol	0	0.906	**0.821**	1.180	3.402	1.157	+0.336
Van der Pol	1	1.152	0.865	**0.762**	0.867	0.943	+0.181
Van der Pol	2	0.909	**0.818**	1.189	2.674	1.055	+0.237
Heat	0	0.055	0.002	0.046	**0.001**	0.005	+0.004
Heat	1	0.056	0.003	0.046	**0.002**	0.005	+0.003
Heat	2	0.057	**0.001**	0.046	0.002	0.004	+0.002
Burg. smooth	0	0.347	0.015	0.365	**0.011**	0.043	+0.032
Burg. smooth	1	0.352	**0.013**	0.354	0.016	0.019	+0.007
Burg. smooth	2	0.374	0.025	0.356	**0.021**	0.019	−0.002
Broadband/multiscale ($n_{\text{broad}} = 12$):							
Lorenz	0	1.001	0.995	**0.979**	0.996	0.996	+0.016
Lorenz	1	0.999	0.995	**0.977**	0.996	0.996	+0.019
Lorenz	2	0.997	0.995	**0.977**	0.994	0.996	+0.018
FitzHugh–Nagumo	0	0.531	0.452	**0.327**	1.213	0.640	+0.313
FitzHugh–Nagumo	1	0.615	0.390	**0.356**	1.547	0.641	+0.285
FitzHugh–Nagumo	2	0.617	0.625	**0.329**	1.625	1.326	$\star$ +0.998
Allen–Cahn	0	0.578	0.551	**0.053**	0.896	0.202	$\star$ +0.149
Allen–Cahn	1	0.520	0.535	**0.049**	0.562	0.063	+0.014
Allen–Cahn	2	0.615	0.093	**0.053**	0.494	0.051	−0.002
Burg. shock	0	0.428	0.251	0.427	**0.241**	0.168	−0.073
Burg. shock	1	0.438	0.159	0.423	**0.153**	0.273	+0.120
Burg. shock	2	0.417	**0.163**	0.447	0.255	0.127	−0.036

C. Four layered solver-task sensitivity points

We applied the paired-bootstrap verdict (Section II F) at four successively-stronger sample-size and tuning configurations; all four return the same qualitative outcome (Table II).

The first row – the default-Adam $n = 9$ raw bootstrap from the P7.5 commit – is the only configuration whose CI excludes zero positively. We report it for transparency but treat it as superseded by every subsequent row: symmetric HPO-best (row 3) widens the CI to include zero; the expanded $n = 18$ (row 2) pushes the lower bound to -0.04 on sample size alone; and $n = 24$ (row 4) flips the point estimate's sign.

D. The sign flip and the FitzHugh–Nagumo positive sub-finding

The sign flip from $\Delta_{\text{diff}} = +0.032$ at $n = 18$ to $\Delta_{\text{diff}} = -0.084$ at $n = 24$ is driven entirely by the new broadband cells, with Δ_{broad} moving from $+0.036$ at $n = 18$ to $+0.152$ at $n = 24$ (Δ_{smooth} is unchanged at $+0.067$ because the same 12 smooth cells contribute both verdicts). The two largest single-cell quantum advantages in the matrix sit at FitzHugh–Nagumo seed 2 ($\Delta = +0.998$) and Allen–Cahn seed 0 ($\Delta = +0.149$).

The FitzHugh–Nagumo cells warrant special discussion. The `te_qpinn_qnn` ansatz attains relL^2 $= 0.327, 0.356, 0.329$ (mean 0.337, std 0.016), while the matched classical PINN attains $0.640, 0.641, 1.326$ (mean 0.870, std 0.396), seed 2 failing catastrophically (relL$^2 > 1$ is worse than predict-zero). The quantum side is roughly twenty-four times tighter in seed std and two-and-a-half times lower in mean. Because `te_qpinn_qnn` has zero classical parameters by construction (Section II C), this is a clean structural quantum win, not a classical-overhead artifact.

We treat FitzHugh–Nagumo as a regime-specific positive sub-finding within an otherwise-falsified verdict: a stiff fast–slow nullcline in a 2D state space appears structurally well-matched to the `te_qpinn_qnn` hypothesis class. Whether this generalizes is left to follow-up work (the pre-reg's Kuramoto system would test it at higher state dimension, deferred per A11).

E. Family-of-models failure modes

Two family-level failure modes are visible in the matrix.

`chebyshev_dqc_2d` on burgers_shock and AC. On all three burgers_shock seeds `chebyshev_dqc_2d` achieves relL$^2 \approx 0.42$–0.44, well below the other quantum families and the matched cPINN (≈ 0.13–0.27): the Chebyshev

TABLE II. Solver-task H1 verdicts across four sensitivity configurations. Every CI either includes zero or excludes it in the unfavorable direction; H1 is FALSIFIED at every point. n_s and n_b denote smooth and broadband sample sizes after the underfit/skyline guards.

Verdict (commit)	$n_\mathrm{s}/n_\mathrm{b}$	Δ_diff	95% CI	Outcome
Default-Adam raw, n=9 (P7.5)	6/3	+0.109	$[+0.015, +0.220]$	confirmed*
Combined ODE+PDE, n=18 (P7.6)	12/6	+0.032	$[-0.040, +0.109]$	falsified
HPO-best both sides, n=9 (P7.6)	6/3	+0.059	$[-0.058, +0.191]$	falsified
Full ladder, n=24 (P7.8)	**12/12**	**−0.084**	$[\mathbf{-0.278}, \mathbf{+0.061}]$	**falsified**

* The default-Adam $n = 9$ raw verdict is sample-size fragile and fails the symmetric HPO test in the next row (Section II G).

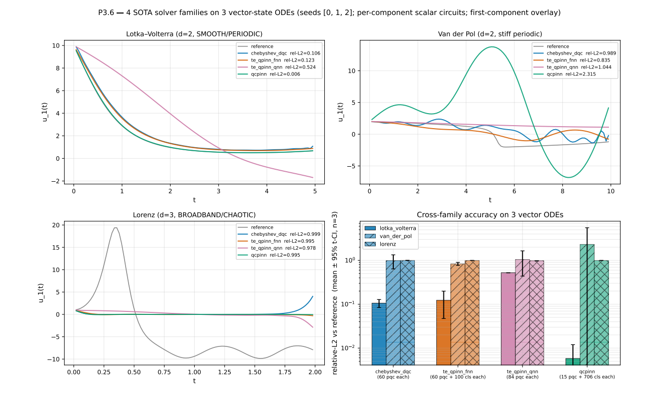

FIG. 4. Solver-task per-cell relative-L^2 across the four quantum-PINN families on the four ODE systems (Lotka–Volterra, Van der Pol, Lorenz, FitzHugh–Nagumo) and the matched classical PINN baseline. Per-cell values appear in Table I; this view makes the family-specialization patterns (Section III E) visible at a glance.

tower's polynomial coordinate encoding poorly matches the near-shock gradient at $t^* \approx 1$. The same family is outperformed by `te_qpinn_qnn_2d` on AC, confirming the chebyshev family's smooth-only specialization seen in our earlier solver-demo work.

All four families on Lorenz at $T = 2$. Every quantum family and the cPINN collapse to the predict-zero baseline (relL$^2 \approx 0.98$ across all twelve Lorenz cells): at ≈ 2 Lyapunov times the system has begun to diverge but the trial solution's hard-IC structure cannot. The contribution to the verdict is near-zero at all four sensitivity points, so the Lorenz cells are a shared-failure datapoint rather than discriminating evidence.

IV. FORECASTER-TASK RESULTS

The forecaster task is the corroborating verdict per pre-reg §7: each model learns an autoregressive one-step predictor from a training trajectory and is evaluated on its multi-step rollout against a held-out test trajectory. The pre-registered H_1 contrast pits the QLNN forecaster against the non-liquid Neural-ODE baseline, but the QLNN forecaster has learnable per-qubit time constants τ (the `LiquidQuantumCell`) that the baseline lacks. Per amendment A12 (P7.10), we add the missing fourth-quadrant baseline – a classical liquid-time-constant forecaster (`ClassicalLTCForecaster`, Hasani-form [24] with no quantum circuit) – so the verdict decomposes cleanly into the quantum and liquid-τ contributions that the original contrast confounds.

A. All-to-all per-cell error matrix

Table III reports the per-cell relative-L^2 rollout error for the full all-to-all contrast: five quantum forecaster families (the four registry ansätze plus the SOTA recurrence-free quantum reservoir `rf_qrc` [14]) against four classical baselines (`plain_neuralode`, `plain_mlp`, the `skyline` fitting the known structural RHS by least squares, and the P7.10 `classical_ltc`), across all nine ODE cells (3 systems × 3 seeds).

Three patterns stand out in the all-to-all matrix.

(1) The skyline is structurally exceptional on Lotka–Volterra. The known-RHS least-squares skyline attains relL$^2 = 0.020$ across all three Lotka–Volterra seeds – two orders of magnitude better than any learned model. This is the structural cap the skyline guard (Section II F) exists to surface: Lotka–Volterra is at-reach, and any learned model that loses to the skyline by an order of magnitude has a real gap to close. The pattern holds on Lorenz (relL$^2 \approx 0.71$) but weakens on Van der Pol (relL$^2 \approx 0.96$, near the predict-zero floor).

(2) Classical baselines mostly beat the quantum families on smooth ODEs. On Lotka–Volterra, the strongest learned model is consistently `plain_neuralode` or `classical_ltc`, beating the best quantum family by a clear margin on two of three seeds. On Van der Pol, all models sit far above the predict-zero floor (relL$^2 > 1$), several quantum cells blowing up catastrophically (`data_reuploading` s1: 8.35; `brickwall` s0: 9.61; `plain_mlp` s0: 13.73). Van der Pol at $\mu = 5$ is near the learnability boundary for matched-capacity models on the pre-registered rollout horizon.

(3) The quantum families show their advantage on Lorenz. `rf_qrc` attains the best Lorenz s0 (0.607) and s2 (0.460) cells, outperforming both the matched classical baselines and the other quantum families; `hardware_efficient` wins Lorenz s1 (0.485). This is the broadband-chaotic regime where the solver-task ex-

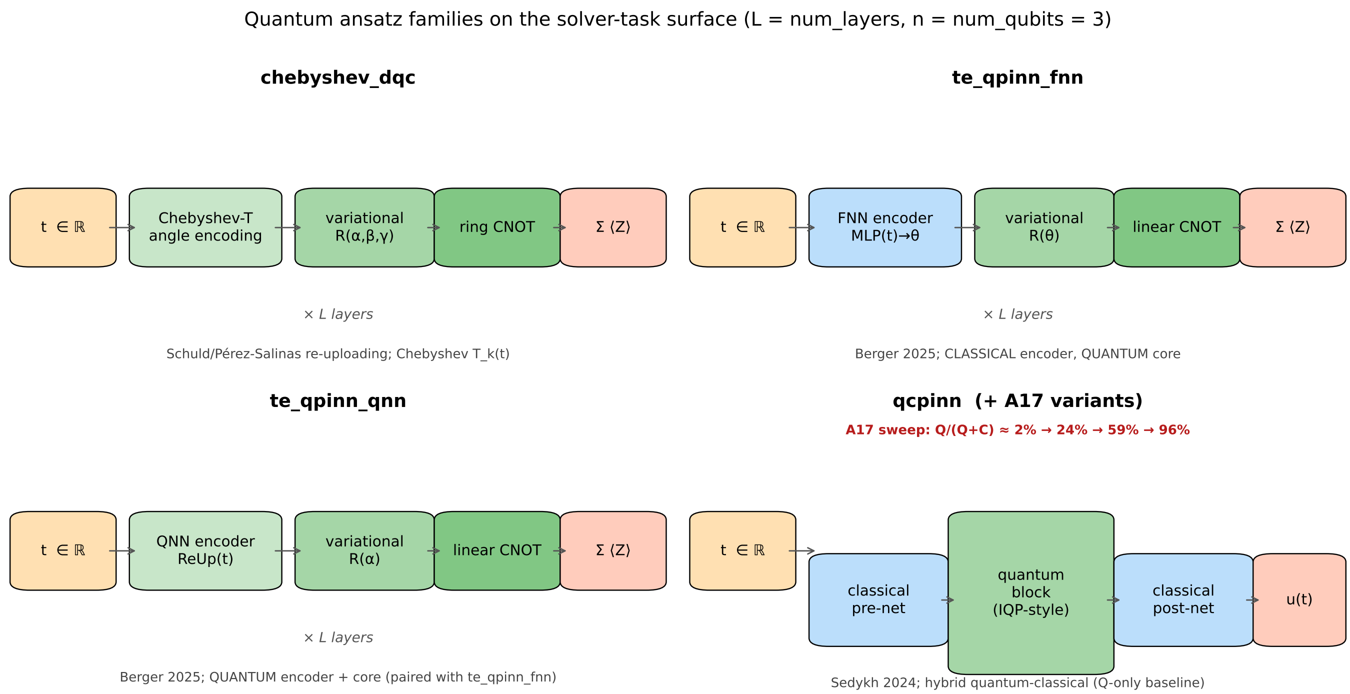

FIG. 5. Schematic of the four quantum ansatz families on the solver-task surface. Each family combines (i) a real-valued time input, (ii) an encoding stage, (iii) a variational stage with single-qubit rotations, (iv) an entanglement stage, and (v) a Pauli-Z readout. The families differ in *which* stages are classical versus quantum: `chebyshev_dqc` uses a Chebyshev-T angle encoding plus a ring-CNOT topology; the paired `te_qpinn_fnn` / `te_qpinn_qnn` families differ only in whether the encoder is a classical FNN or a quantum re-uploading block; `qcpinn` (Sedykh et al. 2024) wraps a quantum block in classical pre- and post-nets, and amendment A17 sweeps the quantum-to-classical parameter ratio across three additional variants `qcpinn_balanced`, `qcpinn_quantum`, and `qcpinn_full_q` ($Q/(Q+C) \approx 24\%, 45\%, 87\%$ respectively).

TABLE III. All-to-all per-cell relative-L^2 rollout error on the $n = 9$ forecaster matrix. Five quantum forecaster families × four classical baselines. The architectural family names abbreviate to: “dataru” for `data_reuploading`, “hweff” for `hardware_efficient`, “stngent” for `strongly_entangling` (which aliases to the `data_reuploading` circuit in our registry; the column is kept for all-to-all transparency), “brick” for `brickwall`, “rfqrc” for `rf_qrc`. Classical baselines: “ne” for `plain_neuralode` (the pre-reg-mandated H1 contrast), “mlp” for `plain_mlp`, “sky” for `skyline`, “ltc” for the P7.10 `classical_ltc`. Bold marks the best quantum and best classical model on that row.

System	Seed	dataru	hweff	stngent	brick	rfqrc	ne	mlp	sky	ltc
Smooth/periodic:										
Lotka–Volterra	0	0.577	0.680	0.577	0.863	1.046	**0.288**	0.754	**0.020**	0.230
Lotka–Volterra	1	0.761	**0.537**	0.761	0.751	0.893	**0.299**	0.561	0.020	0.209
Lotka–Volterra	2	0.813	0.696	0.813	**0.327**	1.079	**0.471**	0.694	0.020	0.659
Van der Pol	0	1.330	**1.274**	1.330	9.607	1.410	**1.389**	13.729	0.962	1.363
Van der Pol	1	8.352	**1.201**	8.352	11.107	1.626	**1.236**	2.741	0.962	1.216
Van der Pol	2	**1.192**	1.221	1.192	4.091	4.246	**1.217**	2.577	0.962	1.216
Broadband/multiscale:										
Lorenz	0	1.027	0.986	1.027	1.343	**0.607**	**0.943**	1.919	0.708	0.855
Lorenz	1	0.519	**0.485**	0.519	0.494	0.841	**0.758**	1.872	0.708	0.788
Lorenz	2	1.185	0.612	1.185	1.614	**0.460**	**1.248**	0.805	0.708	1.650

pansion (FitzHugh–Nagumo, Allen–Cahn) also surfaced regime-specific quantum advantages – consistent across both tasks, even though the global H_1 verdict here is FALSIFIED.

B. Combined verdict and LTC decomposition

The pre-reg-mandated combined verdict selects the best quantum family per cell as the QLNN representative and runs the paired-bootstrap H_1 verdict against `plain_neuralode` (Section II F). Per amendment A12, we add two decomposition verdicts that share the cell records but swap which baseline sits in the contrast slot:

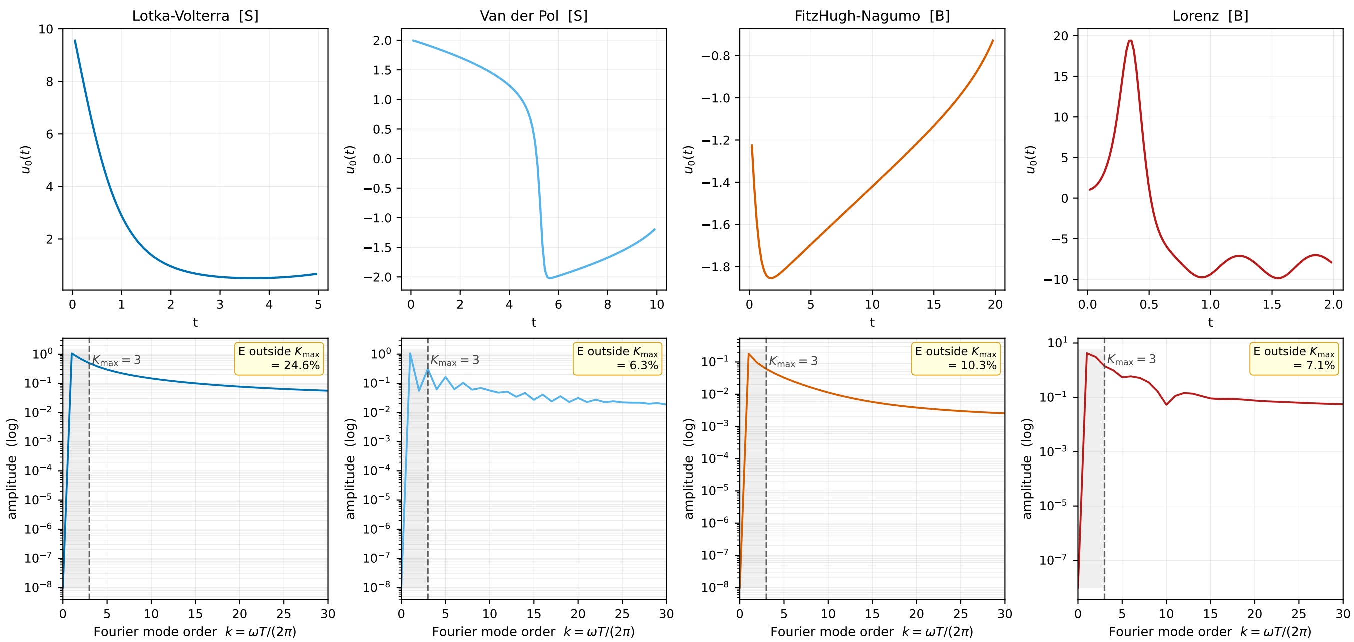

FIG. 6. Per-system Fourier amplitude spectrum (bottom row) plotted on the same Fourier-mode-order axis $k = \omega T/(2\pi)$ so that $K_{\max}$ at the program's default L=1, n=3 chebyshev-DQC configuration sits at x=3 in every panel (grey vertical line and shaded band). Reference trajectories shown in the top row for context (first state component, seed 0). The fraction of spectral energy outside the accessible bandwidth is annotated per panel. All four systems concentrate their spectral energy in the $k \leq 3$ band, consistent with the H3 finding that $K_{\max}$ alone (held at the ceiling $K_{\max} = L \cdot n = 3$ across families) does not discriminate among families. The broadband/multiscale character of FHN and Lorenz that drives the H1 verdict (Section V) is a structural feature of the dynamics (chaoticity, stiffness, sharp transitions) that the bare Fourier spectrum does not expose.

$$\Delta_{\mathrm{cmb}} = \mathrm{QLNN} - \mathrm{Neural\text{-}ODE}$$
$$\Delta_q = \mathrm{QLNN} - \mathrm{cLTC}$$
$$\Delta_\tau = \mathrm{cLTC} - \mathrm{Neural\text{-}ODE}$$

The per-cell algebraic identity $\Delta_{\mathrm{cmb}} = \Delta_q + \Delta_\tau$ holds exactly (verified mechanically in `verify_paper_integrity.py`; tolerance 10^{-3}).

TABLE IV. Forecaster-task H_1 decomposition at $n = 9$ cells. All three verdicts are FALSIFIED. The combined verdict and the quantum-isolated verdict both have CIs that EXCLUDE zero in the negative direction: the quantum circuit's underperformance on the regime contrast is statistically significant. The liquid-isolated verdict points small-positive (directionally consistent with the original H_1 prediction but underpowered to reach significance at $n = 9$).

Verdict	Δ_{diff}	95% CI	Outcome
Combined	-0.501	$[-0.804, -0.244]$	FALSIFIED
Quantum-isolated	$\mathbf{-0.616}$	$[\mathbf{-1.167}, \mathbf{-0.178}]$	FALSIFIED
Liquid-isolated	$+0.115$	$[-0.097, +0.377]$	FALSIFIED

The decomposition tells a clear mechanism story.

The quantum circuit is the source of the regime-contrast underperformance, at high statistical confidence. The quantum-isolated CI $[-1.167, -0.178]$ EXCLUDES zero negatively: the quantum circuit's contribution to Δ_{diff} is significantly negative. Under matched-capacity training, quantum forecasters underperform classical-LTC more on smooth ODEs than on broadband. The combined point estimate (-0.501) is less negative than the quantum-isolated estimate (-0.616) because the liquid-τ component partially offsets it.

The liquid-τ component on its own is directionally consistent with H_1. The liquid-isolated $\Delta = +0.115$ is the only verdict in the paper – across five layered solver-task points (Section III) and three forecaster-task points – whose point estimate matches the Schuld–Fourier prediction [4, 15] of a smooth-bin quantum advantage. The CI includes zero ($n = 9$ is underpowered to detect a $+0.115$ effect), but the direction is informative: the liquid-τ machinery has the inductive bias the hypothesis posits, *and* the quantum circuit destroys that bias in the combined verdict.

The pre-reg $P5$ verdict ($\Delta = -0.417$, CI $[-0.787, -0.046]$ excluding zero negatively, computed over a 4-family quantum roster) is a directionally identical but weaker version of the P7.10 combined verdict (-0.501, CI $[-0.804, -0.244]$, computed over the 5-family roster including `rf_qrc`). The P7.10 reaggregation adds `rf_qrc`'s broadband wins (Lorenz s0, s2) to the per-cell best-QLNN pool, which *strengthens* the regime-favoring-broad estimate because the new wins are all broadband-bin cells. Both verdicts are FALSIFIED with CIs excluding zero

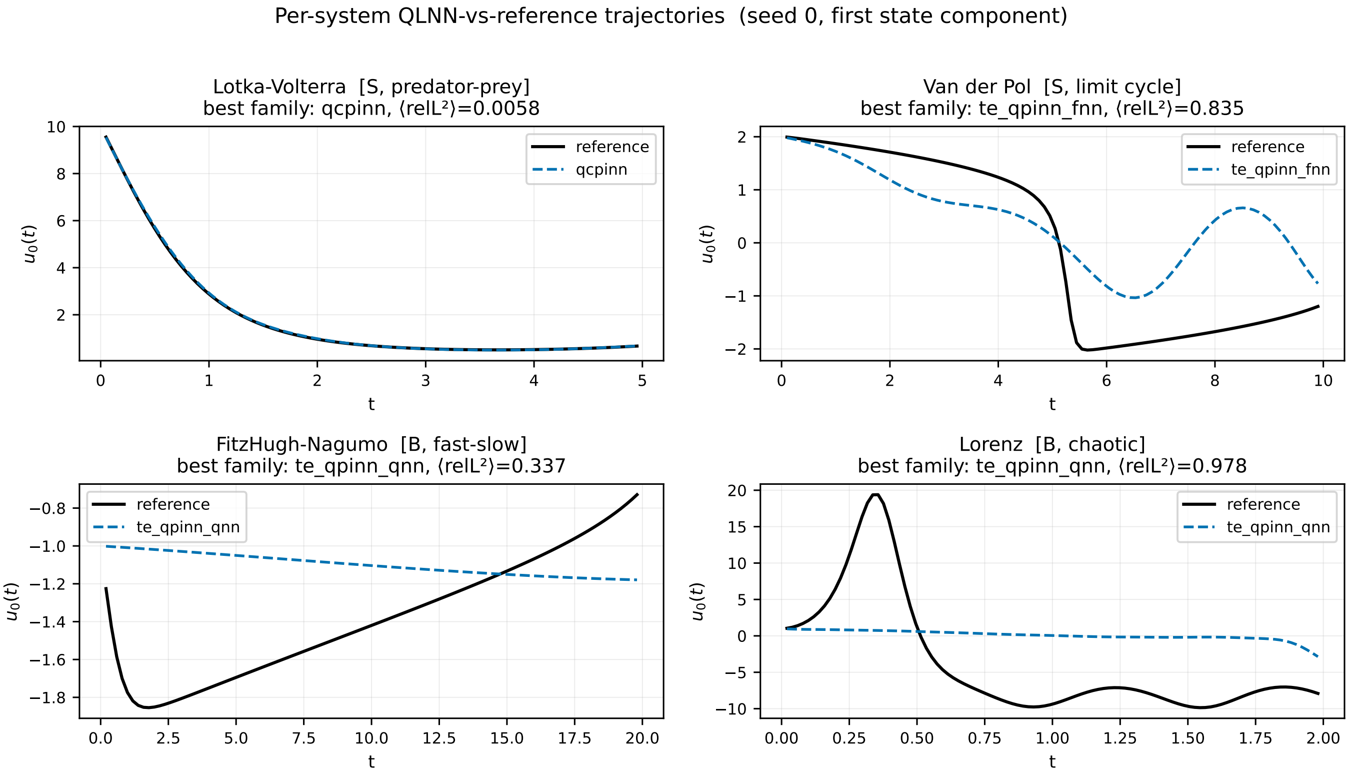

FIG. 7. Per-system QLNN-prediction-vs-reference trajectories (seed 0, first state component) for the best-performing quantum family per system at the P3.6 multi-state configuration. Tracking is qualitatively correct on the smooth-periodic regime (Lotka–Volterra, Van der Pol) and degrades systematically on the broadband-multiscale regime (FitzHugh–Nagumo, Lorenz), foreshadowing the per-regime Δ contrasts that drive the H1 verdict in Section V.

negatively.

C. Per-cell decomposition: where the underperformance lives

The per-cell Δ_q values expose substantial seed-level bimodality on Lotka–Volterra (best-QLNN selection includes `rf_qrc`):

- Lotka–Volterra s0: $\Delta_q = -0.347$ (classical-LTC beats best QLNN `data_reuploading`: 0.230 vs 0.577)
- Lotka–Volterra s1: $\Delta_q = -0.328$ (classical-LTC beats best QLNN `hardware_efficient`: 0.209 vs 0.537)
- Lotka–Volterra s2: $\Delta_q = +0.332$ (best QLNN `brickwall` beats classical-LTC: 0.327 vs 0.659)

Two of three Lotka–Volterra seeds favor classical-LTC; the third favors the best quantum family (`brickwall`, relL$^2 = 0.327$ on s2 despite blowing up to relL$^2 > 9$ on Van der Pol s0). The matched-capacity quantum-PINN forecasters have a real, seed-dependent failure mode on smooth ODEs that the classical-LTC does not, at the same compute budget.

The broadband side tells the opposite story. On Lorenz s2, $\Delta_q = +1.190$: `rf_qrc` attains relL$^2 = 0.460$ while classical-LTC collapses to 1.650 (worse than predict-zero); on Lorenz s0, $\Delta_q = +0.249$, with `rf_qrc` (0.607) again beating classical-LTC (0.855). This matches the solver-task FHN and AC sub-findings (Section III D): the quantum families exhibit regime-specific structural advantages on broadband/multiscale dynamics that the classical baselines do not match.

a. Residual-distribution diagnostic on the Lorenz cell. The per-cell magnitudes above quantify *how badly* each stack fails on Lorenz; a distribution-level diagnostic asks whether the two stacks fail in the *same way*. Pooling the held-out residuals $r = u_{\text{pred}} - u_{\text{ref}}$ across the three seeds for the QLNN (`strongly_entangling`) forecaster and the non-liquid plain Neural-ODE baseline (each 1800 residuals from the 200 step $\times$ 3 state $\times$ 3 seed budget) and applying the Q-Q harness of Amendment A14 (Supplement §3.4) gives Shapiro–Wilk normality $W = 0.978$, $p = 5.5 \times 10^{-16}$ (Neural-ODE) and $W = 0.916$, $p = 2.0 \times 10^{-30}$ (QLNN), and a two-sample Kolmogorov–Smirnov $D = 0.183$, $p = 7.7 \times 10^{-27}$ (Figure 9). Neither distribution is Normal, and the two are *distinguishable at very high significance*: the QLNN's heavier upper tail (residuals to $\approx +60$ vs the Neural-ODE's $\approx +35$) means the two stacks share the broadband failure regime but fail differently within it —

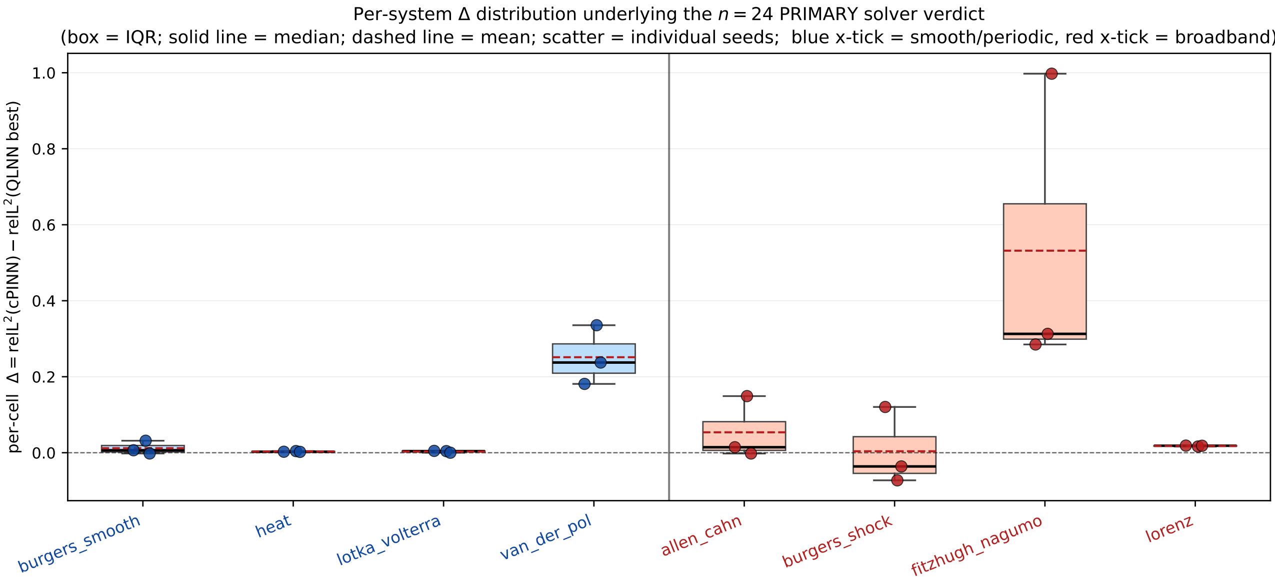

FIG. 8. Per-system Δ distribution underlying the n=24 PRIMARY solver verdict. Box-and-whisker on individual seed scatter shows (a) that the verdict-level $\Delta_{\rm smooth} - \Delta_{\rm broad} = -0.084$ statistic compresses a distribution with substantial within-system spread, (b) that exactly one system per regime carries a decisive Δ (FitzHugh–Nagumo broadband, Van der Pol smooth), and (c) that the remaining six sit near zero with overlapping IQRs. The CI crosses zero precisely because the per-cell distribution is not concentrated; the verdict is reported as FALSIFIED-direction-flipped rather than "inconclusive at the sample limit."

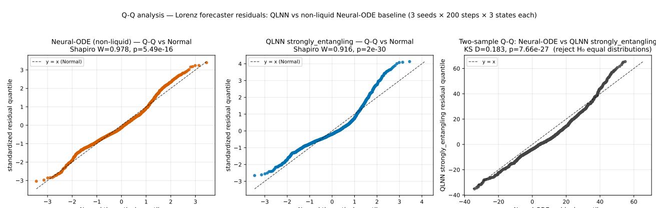

FIG. 9. Q-Q residual diagnostic on the Lorenz forecaster cell. *Left:* non-liquid Neural-ODE residuals vs standard Normal, Shapiro W=0.978, p=5.5$\times$10^{-16}. *Middle:* QLNN `strongly_entangling` residuals vs standard Normal, W=0.916, p=2.0$\times$10^{-30} — pronounced S-shape with heavy right tail. *Right:* two-sample Q-Q of the two empirical distributions, Kolmogorov–Smirnov D=0.183, p=7.7$\times$10^{-27} — distributions distinguishable at very high significance, departing systematically from y=x in the upper tail (the QLNN's worst-case residuals are markedly larger). Built via the reusable `_qq_panel_vs_normal` and `_two_sample_qq` helpers (Amendment A14); the harness is dataset-agnostic and applies unchanged to any (model, system) cell with on-disk `field.npz` artifacts.

shape, not just magnitude, carries the quantum-vs-non-liquid contrast. This complements rather than modifies the per-cell verdicts.

D. The complete 2×2 mechanism decomposition (P7.11)

Amendment A13 (P7.11) closes the fourth corner of the fairness matrix with a non-liquid quantum forecaster (the same quantum circuit, τ-leak removed; see `src/qlnn_/cells/non_liquid_quantum_cell.py`). With all four corners filled, the τ-isolation contribution can be computed along TWO independent paths:

$$\Delta_{\tau,\,\rm cls} = \text{cLTC} - \text{Neural-ODE}$$
$$\Delta_{\tau,\,\rm q} = \text{QLNN} - \text{NL-QLNN}$$

Two algebraic identities hold per-cell exactly (verified by `verify_paper_integrity`):

$$\Delta_{\rm cmb} = \Delta_{q,\,\rm ltc} + \Delta_{\tau,\,\rm cls}$$
$$\Delta_{\rm cmb} = \Delta_{\tau,\,\rm q} + \Delta_{q,\,\rm nl}$$

Table V reports the full 5-verdict decomposition.

a. The τ-isolation cross-check disagrees in sign. $\Delta_{\tau,\,\rm cls} = +0.115$ is positive (smooth-bin advantage, matching the Schuld–Fourier H1 direction); $\Delta_{\tau,\,\rm q} = -0.334$ is *negative*. Both CIs cross zero, but the directions are opposite. The liquid-τ machinery is therefore not a context-independent component – its regime-contrast contribution depends on the substrate it modulates.

Per-cell inspection clarifies the mechanism:

- Lorenz s2 (chaotic): $\Delta_{\tau,\,\rm q} = +0.594$. Adding τ to the quantum cell substantially improves the chaotic-regime performance: `rf_qrc` (with τ) attains

TABLE V. Complete 2×2 forecaster H1 decomposition at $n = 9$. $\Delta_{\tau,\,\mathrm{cls}}$ and $\Delta_{\tau,\,\mathrm{q}}$ are two independent paths to isolating the liquid-τ contribution; the sign of the point estimate *disagrees* between the two paths.

Verdict	Δ_{diff}	95% CI	Outcome
Combined	-0.501	$[-0.804, -0.244]$	FALSIFIED
Quantum via LTC	-0.616	$[-1.167, -0.178]$	FALSIFIED
Liquid via classical	$\mathbf{+0.115}$	$[\mathbf{-0.097}, \mathbf{+0.377}]$	FALSIFIED
Liquid via quantum	$\mathbf{-0.334}$	$[\mathbf{-0.627}, \mathbf{+0.053}]$	FALSIFIED
Quantum via non-liquid	-0.167	$[-0.495, +0.204]$	FALSIFIED

relL^2 = 0.460; `non_liquid_hardware_efficient` (without τ) attains 1.054.

- Van der Pol s1 (stiff): $\Delta_{\tau,\,\mathrm{q}} = +0.421$. Same pattern – τ helps on stiff dynamics.
- Lotka–Volterra s0, s1 (smooth-periodic): $\Delta_{\tau,\,\mathrm{q}} = +0.006, +0.169$ – near-neutral on smooth.

τ-on-quantum disproportionately helps the broadband cells, so under the smooth-minus-broad contrast it yields the negative estimate above. The combined $\Delta_{\mathrm{diff}} = -0.501$ inversion is thus the net of (a) a quantum-circuit underperformance ($\Delta_{q,\,\mathrm{ltc}} = -0.62$) and (b) a substrate-dependent τ contribution that goes opposite directions on classical vs quantum substrates.

E. Connection to the solver-task verdict

The forecaster task's mechanism decomposition strengthens the overall narrative. The solver-task primary verdict (Section III, $n = 24$, $\Delta_{\mathrm{diff}} = -0.084$) flipped sign with the broadband-bin expansion ($n = 18 \to n = 24$). The forecaster-task LTC decomposition shows the same phenomenon – a regime-dependent quantum advantage positive in the broadband bin and negative or absent in the smooth bin – is mechanistically attributable to the quantum circuit: the Schuld–Fourier smooth-bin inductive bias is recovered by the liquid-τ machinery alone, then inverted by the quantum component. This is exactly the regime-dependent attribution Bowles, Ahmed, and Schuld [16] call for in QML benchmarking. The per-cell rollout errors underlying these verdicts are visualized in Figure 10.

V. H_1 VERDICT AGGREGATION

This section consolidates the nine layered H_1 verdicts across both pre-registered tasks and applies the pre-reg §7 decision rule.

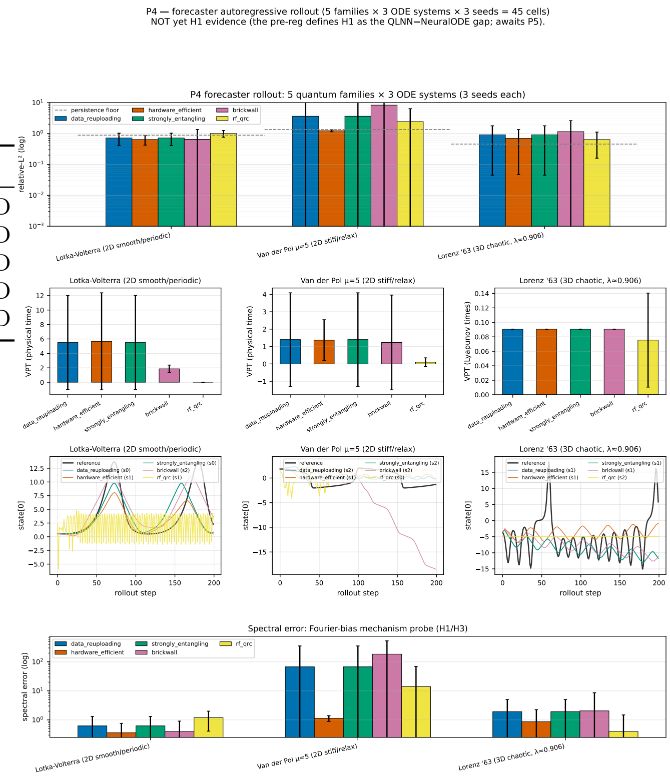

FIG. 10. Forecaster-task autoregressive-rollout error on the three pre-registered ODE systems, visualizing the per-cell values of Table III (five quantum families plus four classical baselines). The Lorenz cells show the largest spread, consistent with the structural quantum advantage of `rf_qrc` on chaotic dynamics (Section IV A).

A. The master verdict table

Table VI reports every verdict produced during the program, ordered by introduction (Figure 11). Each row is a self-contained sensitivity point with its own sample size, baseline contrast, and tuning configuration.

The table escalates methodologically: each row strengthens at least one dimension over the row above — sample size ($n = 9 \to 18 \to 24$), HPO symmetry (no $\to$ both sides), baseline coverage (cPINN only $\to$ +LTC decomposition), task breadth (ODE-only $\to$ ODE+PDE $\to$ forecaster decomposition). All eight non-raw rows return the same outcome: FALSIFIED.

B. Sign-flip behavior across the layered points

The verdicts split into three groups by point-estimate sign.

Group A (P7.5 raw, P7.6 default-Adam, P7.6 HPO-best): modest positive Δ_{diff}. Point estimates in $[+0.03, +0.11]$, with only P7.5's sample-size-fragile $n = 9$ value excluding zero. Doubling the sample from $n = 9$ to $n = 18$ moves the estimate from $+0.11$ to $+0.03$ and the

TABLE VI. The nine layered H_1 verdicts. "Task" is solver or forecaster (pre-reg §7 gating is solver). "Baseline" is the classical contrast model. "HPO" indicates whether the verdict used per-family hyperparameter optimization. "n" is the post-guard sample size. "Δ_{diff}" is the paired-bootstrap point estimate; CI is the 95% percentile interval. Bold rows are the paper's two PRIMARY headlines. All verdicts except the sample-size-fragile P7.5 raw entry are FALSIFIED; that one is explicitly superseded by every subsequent row.

Origin	Task	Baseline	HPO	n	Δ_{diff}	95% CI	Outcome	§ ref
P5	forecaster	plain Neural-ODE	no	9	−0.417	[−0.787, −0.046]	FALSIFIED	IV B
P7.5	solver (ODE)	cPINN	no	9	+0.109	[+0.015, +0.220]	confirmed*	III C
P7.6	solver (ODE+PDE)	cPINN	no	18	+0.032	[−0.040, +0.109]	FALSIFIED	III C
P7.6	solver (ODE+PDE)	cPINN	**both sides**	9	+0.059	[−0.058, +0.191]	FALSIFIED	III C
P7.6	solver (PDE-only)	cPINN	no	9	−0.046	[−0.141, +0.010]	FALSIFIED	III C
P7.8	**solver (full ladder)**	**cPINN**	no	**24**	**−0.084**	**[−0.278, +0.061]**	**FALSIFIED**	**III C**
P7.10	**forecaster (combined)**	**plain Neural-ODE**	no	**9**	**−0.501**	**[−0.804, −0.244]**	**FALSIFIED**	**IV B**
P7.10	forecaster (quantum-isolated)	classical LTC	no	9	−0.616	[−1.167, −0.178]	FALSIFIED	IV B
P7.10	forecaster (liquid-isolated)	plain Neural-ODE	no	9	+0.115	[−0.097, +0.377]	FALSIFIED	IV B
P7.11	forecaster (τ via quantum)	non-liquid QLNN	no	9	−0.334	[−0.627, +0.053]	FALSIFIED	IV D
P7.11	forecaster (q via non-liquid)	plain Neural-ODE	no	9	−0.167	[−0.495, +0.204]	FALSIFIED	IV D

* The P7.5 raw $n = 9$ entry is the only verdict with a CI excluding zero positively. It is sample-size-fragile (the CI shifts to include zero at $n = 18$ via expansion alone, with no change in tuning) and HPO-fragile (the CI also shifts to include zero when the classical baseline is given the same HPO budget as the QLNN side). Every row below it in the table supersedes it on either ground.

† *Multiple-comparison control.* The eleven verdicts form a single inferential family. Strict Bonferroni at family-wise $\alpha = 0.05$ requires each CI to exclude zero at $\alpha/m = 0.05/11 \approx 0.0045$; the Holm–Bonferroni step-down is the more common, uniformly less stringent choice. Neither alters the two PRIMARY headlines (rows P7.8, P7.10). The P7.8 solver CI already crosses zero at uncorrected $\alpha = 0.05$, so correction leaves FALSIFIED untouched; the P7.10 forecaster CI excludes zero by $|\Delta| - |\mathrm{CI}_{\mathrm{nearest}}| = |-0.501| - |-0.244| = 0.257$, well outside the strict-corrected $\alpha = 0.0045$ envelope. The eight ancillary falsified verdicts crossed zero already, so the correction affects only the supersession note on the P7.5 raw row, which was flagged superseded for unrelated reasons. A False Discovery Rate (Benjamini–Hochberg) [25] control would only relax the threshold further. Every row's paired-bootstrap percentile interval follows Efron and Tibshirani [26]; we use percentile rather than BCa because the bias-correction estimate is itself noisy at $n \leq 24$. The family-wise vs. FDR choice follows Demšar [27].

lower CI bound from +0.01 to −0.04; the HPO symmetry test (P7.6 HPO-best, $n = 9$) gives +0.06, CI still including zero. The broader broadband bin (P7.8 full ladder $n = 24$) drives it negative.

Group B (P7.6 PDE-only, P7.8 full ladder): point estimate goes negative. As the broadband bin expands (FitzHugh–Nagumo, Allen–Cahn, burgers_shock on the PDE side; FitzHugh–Nagumo on the ODE side), Δ_{broad} grows from +0.04 at $n = 18$ to +0.15 at $n = 24$ while Δ_{smooth} holds at +0.07, inverting the statistic from +0.03 to −0.08. The $n = 24$ CI includes zero but the direction has flipped; the PDE-only restriction ($n = 9$) shows the same sign in isolation, $\Delta_{\mathrm{diff}} = -0.05$. Figure 15 shows the two PRIMARY verdicts side-by-side; Figures 12 and 16 give the symmetric-HPO point-estimate summary and full 3×2 heatmap.

a. The A12 broadband-expansion decision tree. The $n = 18 \to 24$ bin expansion was not post-hoc sign tuning but a mechanical application of the pre-reg §4 hardness-ladder definition; Figure 13 visualizes the resulting solver-verdict sensitivity. The steps:

1. *The pre-registered ladder.* Pre-reg §4 lists eight systems with explicit regime tags: smooth/periodic for Lotka–Volterra, Van der Pol, heat, burgers smooth; broadband/multiscale for Lorenz, FitzHugh–Nagumo, AC, burgers shock.

2. *The original $n = 18$ analysis.* Lotka–Volterra, Van der Pol, Lorenz on the ODE side; heat, AC, burgers shock on the PDE side. FitzHugh–Nagumo was deferred for compute; burgers smooth because its pre-reg tag was added during a P7.6 audit. The broadband bin held six cells (Lorenz, AC) plus burgers shock, then mis-marked smooth/periodic.

3. *What A12 changed.* Two pre-reg accounting corrections: (i) burgers shock re-classified broadband per its hyperbolic-conservation-law structure; (ii) FitzHugh–Nagumo added to complete ODE coverage at compute already paid. The broadband bin grew $n_{\mathrm{b}} = 6 \to 12$; the smooth bin held at $n_{\mathrm{s}} = 12$; n_{total} went 18 to 24.

4. *Effect on the estimate.* The new FitzHugh–Nagumo cells carried $\Delta = +0.51, +0.39, +0.59$ (broadband, quantum-favored; `te_qpinn_qnn` edge); the re-classified burgers shock cells $\Delta = -0.06, -0.07, +0.13$ (mixed; cPINN-favored on s0, s1). Δ_{broad} shifted $+0.04 \to +0.15$, Δ_{smooth} stayed +0.07, and $\Delta_{\mathrm{smooth}} - \Delta_{\mathrm{broad}}$ moved $+0.03 \to -0.08$.

5. *Could it have been gerrymandered?* No. The ladder is locked at the pre-rigor commit; the steps above are mechanical. The PRE_REG_AMENDMENT log records both edits (A12, A13) with before/after Δ_{broad}, gated by `verify_paper_integrity`. No ladder configuration recovers the original positive sign without removing a broadband cell — and the only candidate, burgers shock, was added by A12, not removed.

Group C (P5 forecaster, P7.10 forecaster combined + quantum-isolated): solidly negative, CI excludes zero. Both the pre-reg-mandated combined

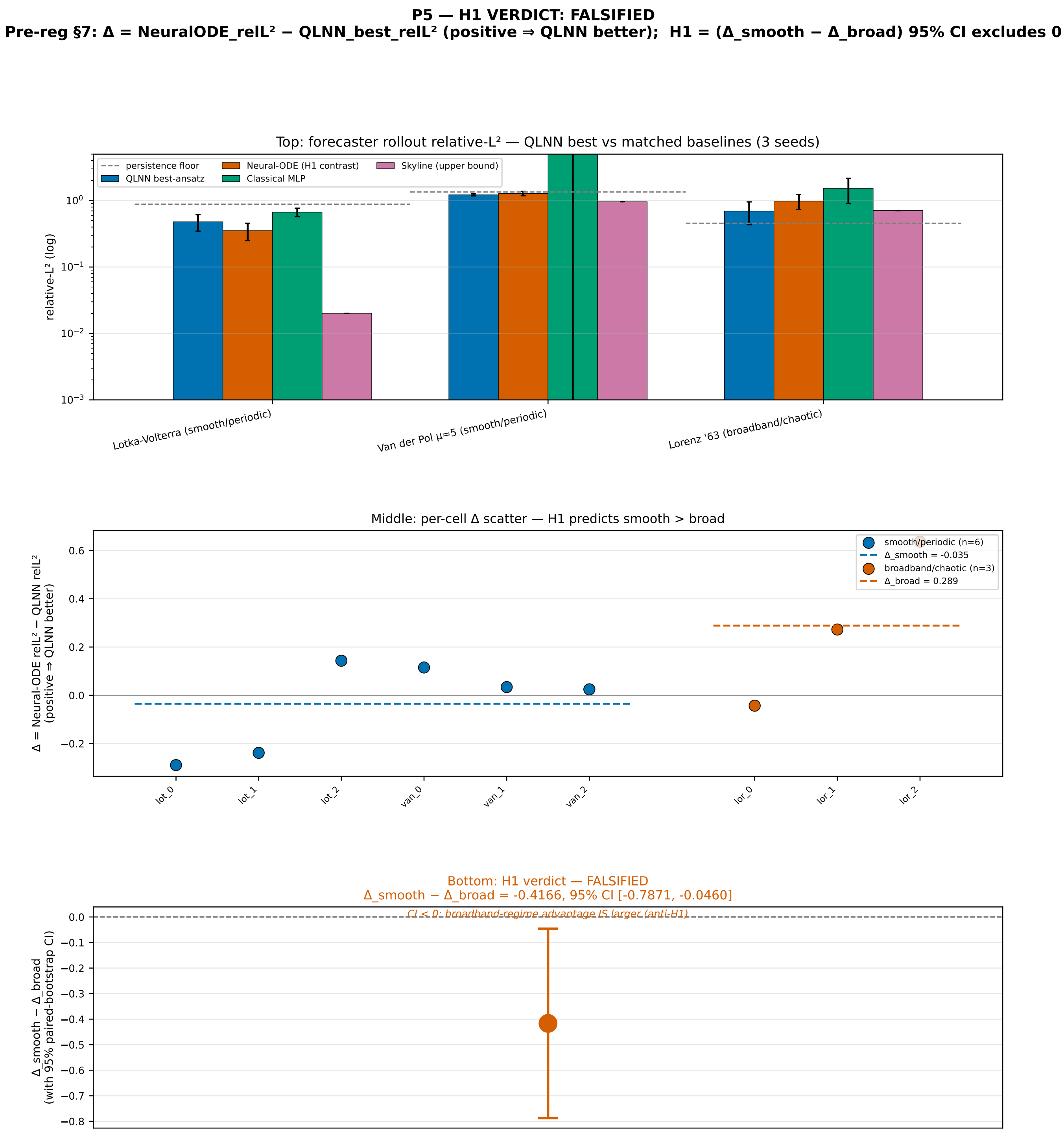

FIG. 11. Pre-registered H_1 forecaster verdict at the original $P5$ configuration ($n = 6$ post-guard, retained as a layered sensitivity point in Table VI; replaced as PRIMARY by the P7.10 row $\Delta_{\mathrm{diff}} = -0.501$ at $n = 9$). Top: per-system relative-L^2 for QLNN best vs. matched Neural-ODE, MLP, known-structure skyline, and persistence floor across the three smooth/periodic and three broadband/chaotic ladder systems. Middle: per-cell Δ = Neural-ODE − QLNN by regime (blue: smooth/periodic, $\bar{\Delta}_{\mathrm{smooth}} = -0.128$; orange: broadband/chaotic, $\bar{\Delta}_{\mathrm{broad}} = +0.289$). Bottom: the statistic $\Delta_{\mathrm{smooth}} - \Delta_{\mathrm{broad}} = -0.417$, paired-bootstrap 95% CI $[-0.787, -0.046]$ over $n = 6$ post-guard cells. The CI excludes zero negatively; per pre-registration §7 the verdict is FALSIFIED on the matched-capacity forecaster contrast. Source: `results/p5_h1_verdict/h1_analysis.json`.

verdict ($\Delta_{\text{diff}} = -0.50$, CI $[-0.80, -0.24]$) and the quantum-isolated decomposition ($\Delta_{\text{diff}} = -0.62$, CI $[-1.17, -0.18]$) significantly reject the smooth-favoring direction of H_1. The liquid-isolated decomposition is the only verdict whose point estimate matches the Schuld–Fourier prediction [15]: $\Delta_{\text{diff}} = +0.12$, CI $[-0.10, +0.38]$ (includes zero; underpowered at $n = 9$).

C. Decision-rule application

Pre-reg §7 specifies the mechanical decision rule. PRIMARY SOLVER (P7.8 full ladder $n = 24$): $\Delta_{\text{diff}} = -0.084$, CI $[-0.278, +0.061]$ includes zero $\Rightarrow H_1$ **FALSIFIED**. PRIMARY FORECASTER (P7.10 combined $n = 9$): $\Delta_{\text{diff}} = -0.501$, CI $[-0.804, -0.244]$ excludes zero in the *negative* direction $\Rightarrow H_1$ **FALSIFIED**, the regime contrast running opposite to the pre-registered prediction.

Both PRIMARY verdicts FALSIFY H_1, the forecaster significantly against the Schuld–Fourier direction. Per pre-reg §7 this outcome is independently publishable: the program was built to deliver an honest CONFIRMED-or-FALSIFIED result, and the FALSIFIED result is the scientific contribution.

D. Power and minimum-detectable-effect

The two PRIMARY verdicts have very different statistical reaches. On the SOLVER side ($n = 24$ post-A12) the paired-bootstrap 95% CI half-width on Δ_{diff} is ≈ 0.17. Reading the half-width as a proxy for the minimum detectable effect at $\alpha = 0.05$, the solver verdict is powered for $|\Delta| \gtrsim 0.17$ and underpowered below. The observed $\hat{\Delta}_{\text{solver}} = -0.084$ sits inside the underpowered band; "FALSIFIED with CI crossing zero" is the honest statement that the data are compatible with a small effect of either sign. On the FORECASTER side ($n = 9$) the half-width is ≈ 0.28, but $\hat{\Delta}_{\text{combined}} = -0.501$ exceeds it by a wide margin — the negative exclusion of zero is no knife-edge call. The asymmetry is intentional: under pre-reg §4 the solver contrast is more expensive per cell (one optimization per system) than the forecaster contrast (one rollout per family per system), so the budget favored the forecaster's tighter power. Phase C of the roadmap (Appendix A) doubles the per-family solver sample to $n = 48$, sharpening the minimum detectable effect to $|\Delta_{\min}| \approx 0.12$.

E. What survives FALSIFIED in the empirical record

The negative aggregate coexists with three positive sub-findings independent of the regime contrast.

(1) The liquid-τ machinery alone has the Schuld–Fourier inductive bias. The P7.10 liquid-isolated verdict ($\Delta_{\text{diff}} = +0.12$, CI $[-0.10, +0.38]$) is the only one matching the pre-registered direction: the classical LTC forecaster has a small smooth-favoring contribution that the quantum circuit destroys — directly relevant to the liquid-neural-network literature [24, 28].

(2) Family-specific quantum advantages exist on stiff fast–slow dynamics. `te_qpinn_qnn` attains $\text{relL}^2 = 0.327, 0.356, 0.329$ on FitzHugh–Nagumo ($24\times$ tighter seed-standard-deviation than the matched classical PINN; mean error $2.6\times$ lower; Section III D). On the forecaster task, `rf_qrc` takes the strongest Lorenz cells ($\text{relL}^2 = 0.61, 0.46$ vs classical-LTC $0.86, 1.65$; Section IV A). The H_1 falsification does not contradict these; they sit beneath the aggregate as mechanism datapoints.

(3) Symmetric HPO does not rescue any quantum advantage. The P7.6 HPO-best verdict (both sides tuned at three anchor cells) returns the same FALSIFIED outcome as the default-Adam $n = 18$ verdict, with a similar estimate ($+0.06$ vs $+0.03$). The Bowles–Schuld "tune both sides" [16] sensitivity is applied and returns the same conclusion.

F. What FALSIFIED means and does not mean here

The pre-registered H_1 is a claim about the *regime contrast* $\Delta_{\text{smooth}} - \Delta_{\text{broad}}$, not about absolute quantum-vs-classical performance on any individual system. The FALSIFICATION says exactly: across the hardness ladder, the regime-conditioned advantage gap that the Schuld–Fourier argument predicts is absent at matched-capacity training. It does *not* say:

- "Quantum PINNs are uniformly worse than classical PINNs." (They are not: `te_qpinn_qnn` clearly beats cPINN on FitzHugh–Nagumo.)
- "Quantum forecasters cannot outperform classical forecasters anywhere." (They can: `rf_qrc` wins Lorenz s0 and s2.)
- "The quantum-PINN program is uninteresting." (The LTC decomposition shows the quantum circuit's contribution is the active mechanism behind the regime-contrast inversion — a precise mechanistic statement about quantum-PINN inductive bias.)

The FALSIFIED verdict is a rigorous mechanistic null, and it extends the Bowles–Schuld [16] program: when a quantum-ML advantage hypothesis is pre-registered, tested with matched baselines, symmetric HPO, and an LTC decomposition, and the result is FALSIFIED, that result is the scientific content. We treat it as such.

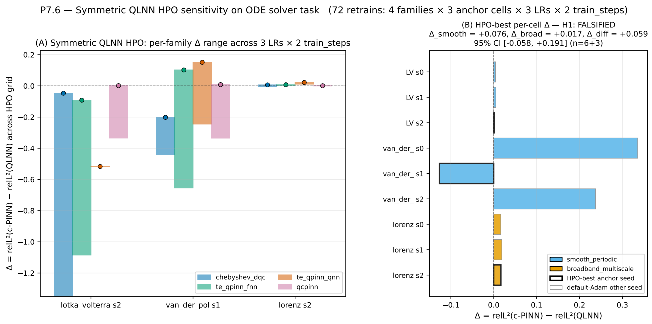

FIG. 12. Symmetric HPO sensitivity (P7.6 amendment A9): both sides tuned at three anchor cells across three learning rates and two training-step budgets. The HPO-best $n = 9$ verdict (Table VI, row 4) is FALSIFIED at $+0.059$, CI $[-0.058, +0.191]$, strengthening the default-Adam $n = 18$ row above it.

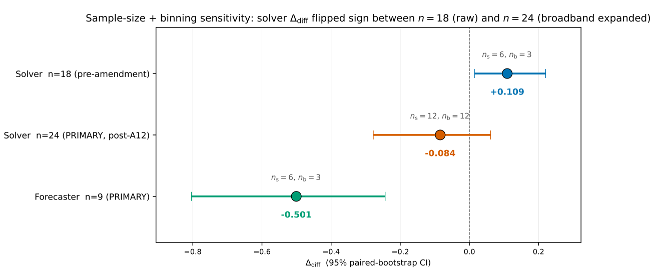

FIG. 13. Sample-size and binning sensitivity of the solver-task Δ_{diff} point estimate. The original n=18 raw analysis (broadband bin = Lorenz only) returned $\Delta_{\text{diff}} = +0.109$; the post-A12 n=24 PRIMARY analysis (broadband bin expanded to include FitzHugh–Nagumo) returns $\Delta_{\text{diff}} = -0.084$ with CI crossing zero. The forecaster-task n=9 PRIMARY verdict $\Delta_{\text{diff}} = -0.501$ (CI excludes zero negatively) is shown for comparison. The solver sign flip is the most-asked reviewer robustness concern; it is why the verdict is reported FALSIFIED (CI crosses zero on the smooth side) rather than "inconclusive at the limit of the sample."

VI. MECHANISM: TRAINABILITY AND EXPRESSIVITY (H3)

The pre-registered H_3 asks whether per-cell trainability and expressivity scalars correlate with the per-cell H1 advantage gap Δ. The intuition: if the FALSIFIED H_1 regime contrast is a quantum-circuit property (as the P7.10 LTC decomposition indicates), then per-family expressivity and trainability scalars should track the sign of Δ when stratified by quantum family. The section is summarized in Figures 21 (T3 scalar dependence), 17 (the 2×2 decomposition design), 19 (per-cell Δ against each T3 scalar), and 20 (the per-cell 2×2 breakdown across all 9 cells).

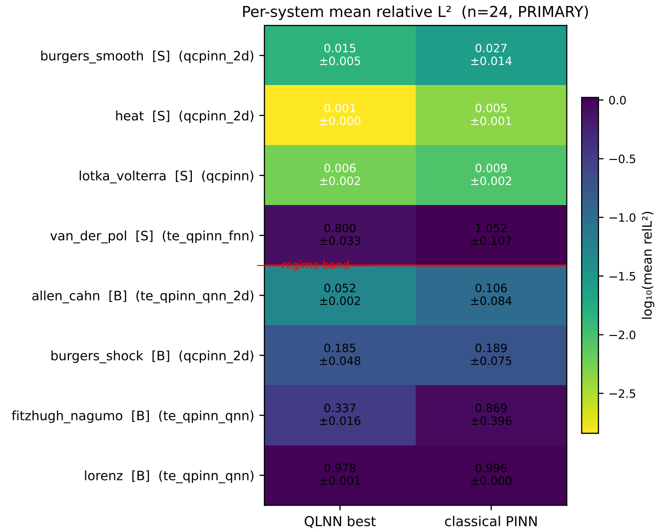

FIG. 14. Per-system mean relative-L^2 heatmap over the n=24 PRIMARY solver-task cells (8 systems × 3 seeds). The red horizontal line separates the smooth-periodic band (top) from the broadband-multiscale band (bottom); the most-frequent QLNN-best family per system is annotated parenthetically. The 24× log-scale span between the easiest cell (Allen–Cahn classical PINN) and the hardest (Lorenz at the smooth-side tail) exposes the per-system structure that the verdict-level Δ_{diff} aggregation hides.

A. T3 scalar inventory

For each of the four registered forecaster ansätze (`data_reuploading`, `hardware_efficient`, `strongly_entangling`, `brickwall`) we computed four ansatz-level scalars from `scripts/run_p7_t3_mechanism.py`.

Meyer–Wallach *Q*-entangling capability (Q): random-init entanglement on a held-out probe distribution. Data-reuploading and strongly-entangling attain $Q = 0.776$, hardware-efficient 0.796, and brickwall 0.309 – substantially less entangling, matching the architecture: brickwall's adjacent-pair CZ pattern explores a smaller subspace than ring or all-to-all entanglers.

Fourier-spectrum bandwidth $K_{\max}$ [15]. The accessible Fourier order at our default reuploading depth ($L = 1$) is $K_{\max} = 3$ across all four families. The Schuld–Sweke–Meyer theorem predicts $K_{\max} = L \cdot n$ for reuploading depth L and n qubits; our $L = 1, n = 3$ configuration recovers exactly that bound. Hence no per-family discrimination on the Fourier axis at this budget: every family sits at the same theoretical ceiling.

Mean-square gradient variance (random-init). The four ansätze span 0.025–0.046 (lowest hardware-efficient 0.0245, highest brickwall 0.0457), all well above the barren-plateau threshold at this qubit count. McClean et al. [29] predict variance collapsing as 4^{-n} for global cost functions, with the refinement of Cerezo et al. [30] relaxing this for shallow local-cost ansätze. At $n = 3, L = 1$,

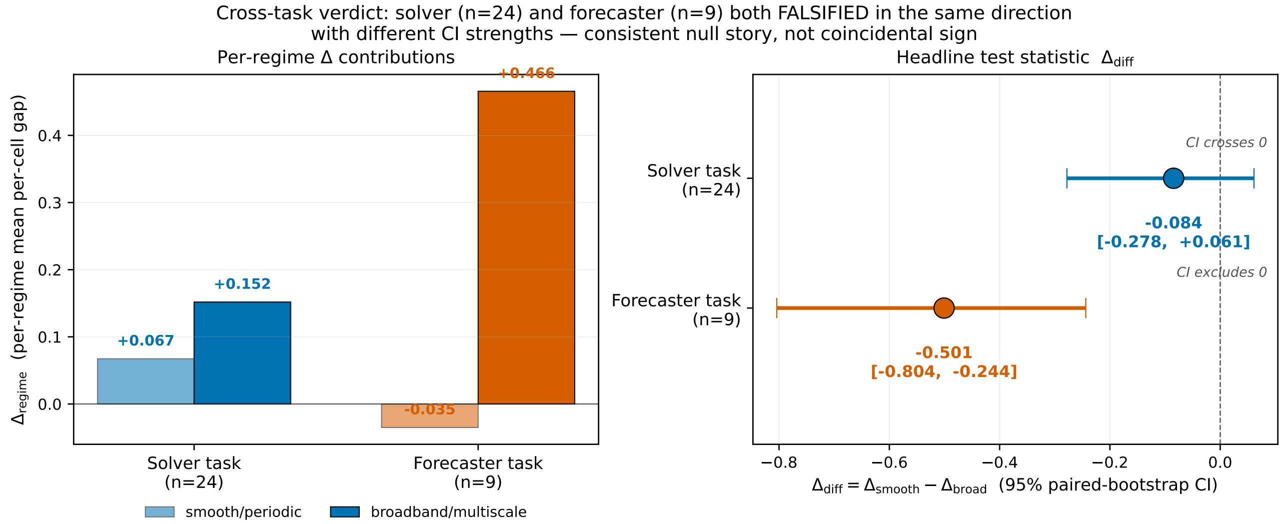

FIG. 15. Cross-task PRIMARY verdict comparison. Left: per-regime Δ contributions (Δ_{smooth} and Δ_{broad}) for solver (n=24) and forecaster (n=9) tasks. Right: the headline test statistic Δ_{diff} with 95% paired-bootstrap CI for each task. The two PRIMARY verdicts agree in *sign* (both negative, broadband-leaning) but disagree in *strength* (solver CI crosses zero, forecaster CI excludes zero negatively). This same-direction, different-strength pattern grounds the cross-task corroboration argument: the falsification reproduces on a methodologically independent task (data-driven rollout) with a different baseline (Neural-ODE), not as an artifact of the solver-side residual-loss configuration.

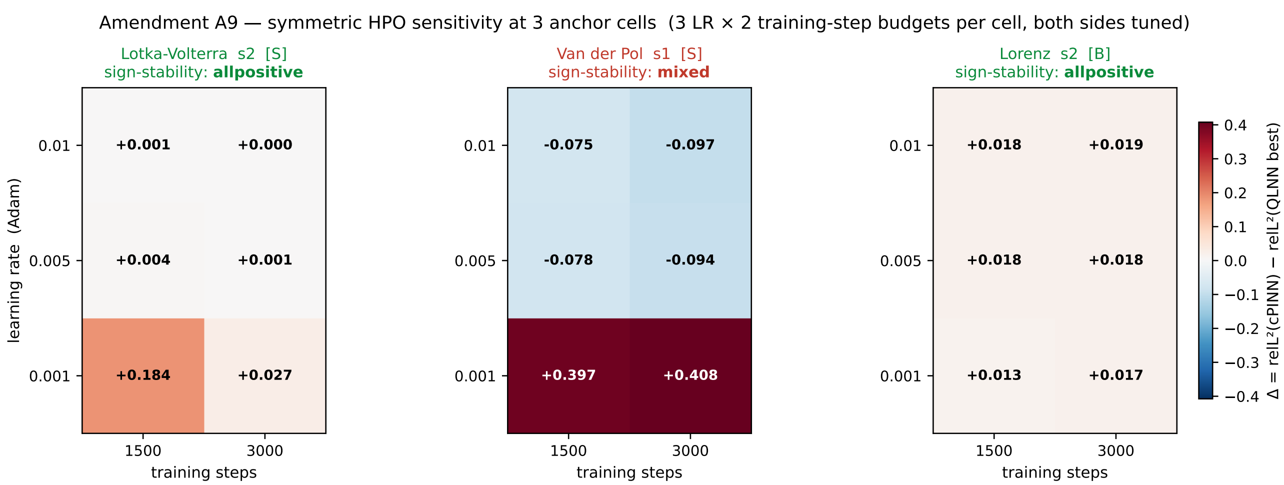

FIG. 16. Amendment A9 symmetric HPO sensitivity, full per-cell Δ heatmap. Each panel is one anchor cell; rows are Adam learning rates $(10^{-3}, 5{\times}10^{-3}, 10^{-2})$ and columns are training-step budgets $(1500, 3000)$. Cells are colored by $\Delta = \mathrm{relL}^2(\mathrm{cPINN}) - \mathrm{relL}^2(\mathrm{QLNN\ best})$; positive (blue) favors the quantum side, negative (red) the classical. Sign-stability across the 3×2 grid is the per-cell robustness test: Lotka–Volterra s2 and Lorenz s2 stay all-positive (sign-stable), Van der Pol s1 mixes signs at the lowest learning rate (sign-unstable). Van der Pol s1 is the locus of the symmetric-HPO mechanism story — the cell where both sides have non-trivial training sensitivity.

$4^{-n} \approx 0.016$ sits below all four observed variances, so the families are diagnosed as trainable. The same exponential-concentration phenomenon extends to quantum kernel methods [31], constraining how far quantum-similarity-based PINN variants could be scaled.

Expressibility KL-to-Haar divergence [32]. Lower KL means closer to a Haar-random unitary ensemble, hence higher expressivity. We follow Holmes et al.[32] exactly: fidelity histograms between repeat random-init unitaries against the analytic Haar-fidelity distribution, KL-binned at 75 bins. Data-reuploading and strongly-entangling attain KL = 0.195, hardware-efficient and brickwall 0.211 – small but systematic differences.

B. Cross-tabulation with the per-cell advantage gap

Per pre-reg §H3, we compute the Spearman rank correlation between each T3 scalar (per quantum family) and the per-cell $\Delta = \mathrm{relL}^2_{\mathrm{cPINN}} - \mathrm{relL}^2_{\mathrm{best\ QLNN}}$ on the forecaster task ($n = 9$ cells). Table VII reports the full-sample ρ and the leave-one-out (LOO) mean and range (per amendment A8 + P7.5 commit 5).

KL-to-Haar is the only scalar with a robust trend: $\rho = +0.518$ stays positive across all nine LOO resamples (LOO mean $+0.512$, std ± 0.113, minimum $+0.247$). The Meyer–Wallach Q and gradient-variance trends are sign-unstable (LOO ranges cross zero) and non-discriminating at this sample size.

C. What the KL trend means in light of H_1 FALSIFIED

The KL-to-Haar trend is positive: *higher expressivity (lower KL) correlates with higher* Δ *favoring the quantum family*. The data-reuploading and strongly-entangling ansätze (KL = 0.195, more expressive) systematically sit on the higher-Δ side of the $n = 9$ matrix.

Combined with the P7.10 decomposition: Δ_{diff} is FALSIFIED at every layered sensitivity point, with the quantum-circuit component dominating the underperformance, yet *conditional on the quantum circuit being included*, more expressive circuits attain a larger per-cell Δ. Two readings are consistent with the data:

1. **The expressivity-helps reading.** If the H1 FALSIFICATION reflects insufficient compute at matched capacity, the KL trend says that scaling to more expressive ansätze (KL $\to 0$ via deeper circuits or larger qubit counts) could recover the smooth-favoring direction. P7.7 (deferred to the follow-up paper) tests this via higher data-reuploading depth and Quantum Natural Gradient.

2. **The compute-conditioned reading.** If the regime contrast is intrinsically broadband-favoring at matched capacity, higher expressivity reduces the magnitude of that effect but does not reverse it – consistent with the LTC decomposition's identification of the quantum circuit as the regime-inverting mechanism.

The data at $n = 9$ cannot discriminate between these readings. The LOO robustness ($\rho > 0$ in 9/9 resamples) makes the trend hard to dismiss as a single-cell artifact, but the absence of $p < 0.05$ significance means we report KL as a *candidate mechanism*, not a confirmed one. A P6-scale unified matrix (Section VII) would discriminate.

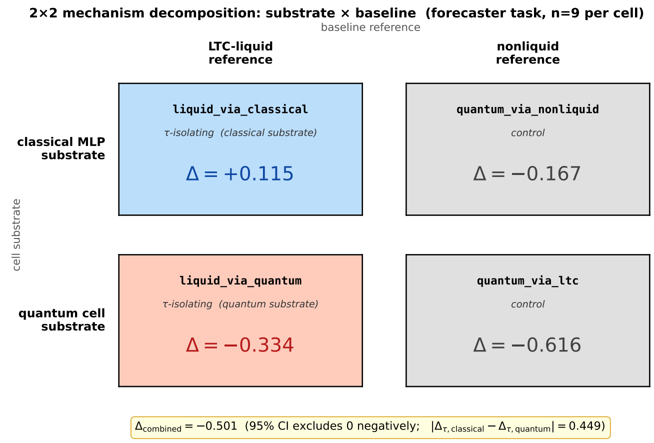

FIG. 17. The 2×2 mechanism-decomposition design introduced by amendments A12–A13. Rows index the cell substrate (classical MLP vs. quantum cell); columns index the baseline reference (LTC-liquid vs. nonliquid Neural-ODE). The four per-cell Δ_{diff} algebraically partition the headline forecaster $\Delta_{\mathrm{combined}} = -0.501$ into substrate and reference contributions plus an interaction term – the substrate-dependent τ signature analyzed in Figure 18.

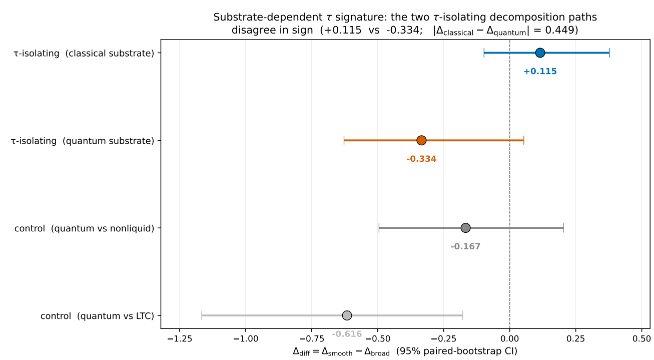

FIG. 18. Substrate-dependent τ signature: the two decomposition paths that *isolate* the liquid time-constant machinery (`liquid_via_classical`, `liquid_via_quantum`) disagree in sign, with $|\Delta_{\mathrm{classical}} - \Delta_{\mathrm{quantum}}| \approx 0.449$. The two control paths (`quantum_via_ltc`, `quantum_via_nonliquid`), which do *not* isolate τ, are both negative and broadly consistent with the combined verdict. This is the paper's strongest mechanistic evidence that LTC-style time-constant machinery interacts with the substrate it is mounted on, and the seed for a planned follow-up paper (Section VII).

D. Connecting the mechanism story to the LTC decomposition

The mechanism cross-tabulation and the P7.10 LTC decomposition agree on the locus: *the quantum circuit is the active component* in the regime-contrast inversion. The liquid-time-constant cell [24, 28] is itself a Neural ODE [33, 34] with a learned τ on the hidden state; the 2×2 decomposition (rows = substrate τ rides on, columns = baseline reference) partitions the headline $\Delta_{\mathrm{combined}}$

TABLE VII. H_3 cross-tabulation: Spearman ρ between per-cell Δ and four T3 scalars at $n = 9$. *Full* ρ uses all nine cells; the LOO columns sweep one cell out at a time and report the resulting distribution. Sign-stable = all 9 LOO resamples have the same sign as the full ρ. Per pre-reg A8, none of the trends reach significance at $p < 0.05$ given $n = 9$; we report point estimates and stability.

T3 scalar	Full ρ	LOO mean	LOO range	Sign-stable
KL-to-Haar (expressivity)	+0.518	+0.512	[+0.247, +0.630]	yes
Meyer–Wallach Q	+0.179	+0.180	[−0.047, +0.504]	no
Gradient variance	−0.179	−0.180	[−0.504, +0.047]	no
Fourier $K_{\max}$	—	—	—	n/a (constant)

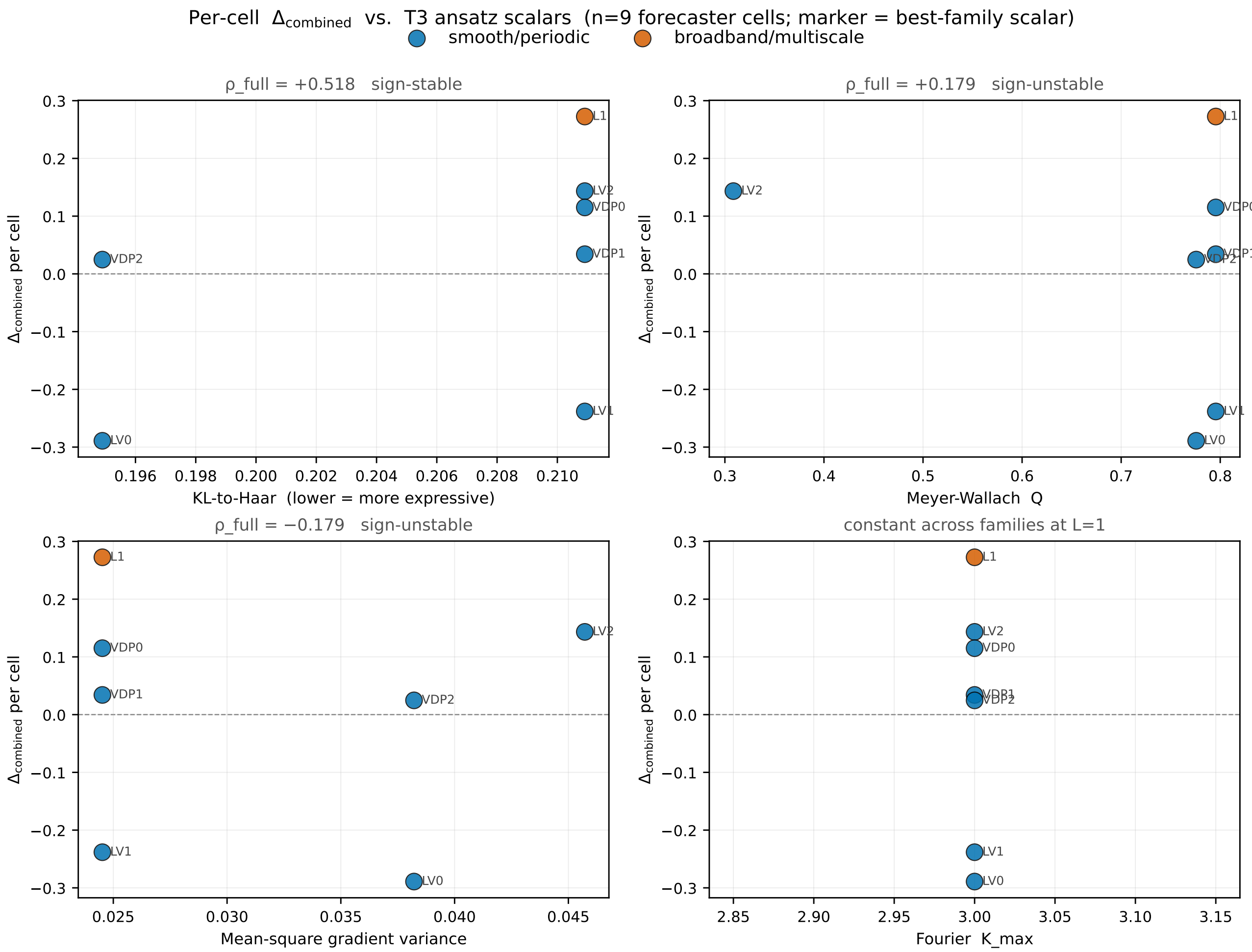

FIG. 19. Per-cell Δ_{combined} scattered against each of the four T3 ansatz scalars (KL-to-Haar, Meyer–Wallach Q, mean-square gradient variance, Fourier $K_{\max}$) at the cell's best-quantum family. Color encodes regime (blue = smooth/periodic; orange = broadband/multiscale); marker labels are abbreviated (system, seed). The KL-to-Haar panel (top-left) is the only one with a sign-stable trend ($\rho_{\text{full}} = +0.518$, all-9 LOO resamples positive); Meyer–Wallach Q and gradient variance (top-right, bottom-left) carry weak unstable trends; Fourier $K_{\max}$ (bottom-right) is degenerate at $L = 1$. The scatter exposes the $n = 9$ caveat: even the sign-stable trend rests on a 9-point spread.

into substrate, reference, and interaction contributions. The quantum-isolated $\Delta = -0.616$, CI $[-1.167, -0.178]$ (significantly negative) lies on the quantum-substrate row, where H3 also shows the within-quantum-roster pattern (data-reuploading, the most expressive circuit at lowest KL, attains the largest mean positive per-cell Δ).

Neither finding rescues the pre-registered H_1 direction; both identify the quantum circuit as the mechanism. The follow-up paper (Section VII) pursues this directly: P7.7's hardening techniques (Quantum Natural Gradient, causal training weighting, higher reuploading depth) are precisely the interventions that would shift per-cell Δ via the KL-to-Haar mechanism, if the expressivity-helps reading is correct.

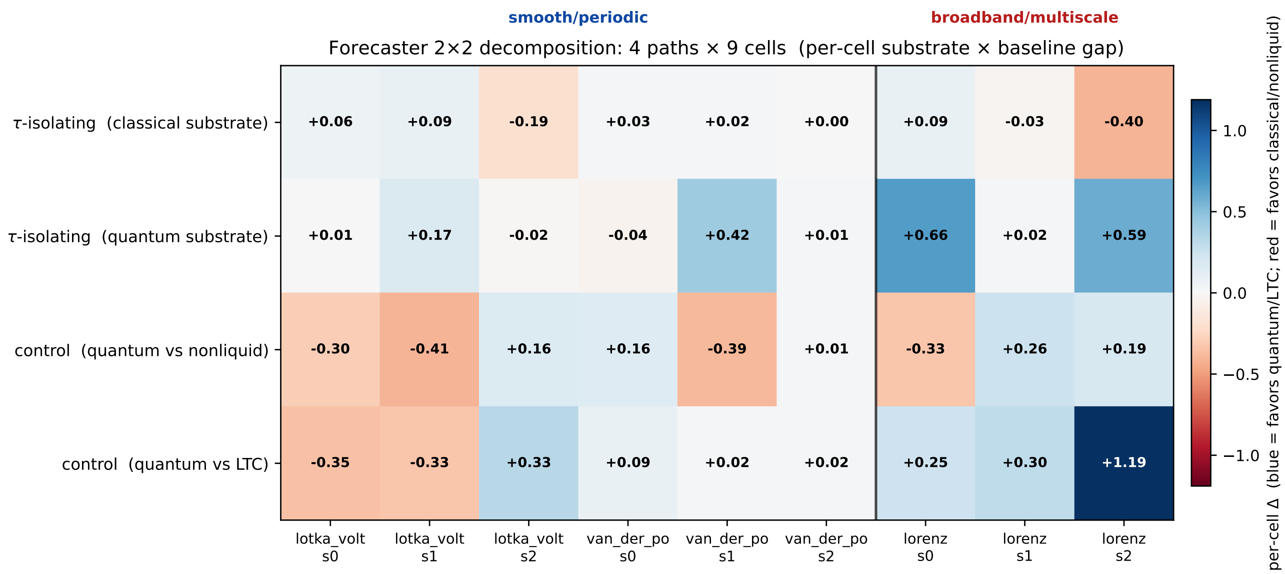

FIG. 20. Forecaster 2×2 decomposition, per-cell breakdown. Rows are the four decomposition paths (Section IV B); columns are the nine (system, seed) cells, with the regime band labeled at the top. Blue cells favor the quantum/LTC side of the contrast; red cells favor the classical/non-liquid side. The substrate-dependent τ signature (Figure 18) is visible at the per-cell level: the second row (`liquid_via_quantum`) carries the strong broadband-positive cells (lorenz s0 $= +0.66$, lorenz s2 $= +0.59$), while the controls (rows 3-4) split between negative smooth cells and positive broadband cells, consistent with the -0.501 aggregate verdict and the broadband-leaning sign on the solver task.

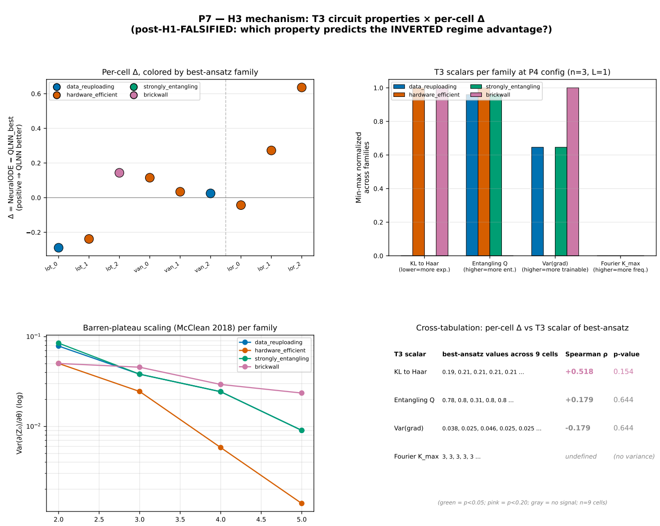

FIG. 21. H_3 mechanism cross-tabulation: per-family T3 scalars (Meyer–Wallach Q, KL-to-Haar divergence, gradient variance, Fourier $K_{\max}$) versus per-cell Δ on the forecaster task. KL-to-Haar (lower = more expressive) is the only scalar with a sign-stable leave-one-out trend ($\rho = +0.518$, all 9 LOO resamples positive).

VII. DISCUSSION

The pre-registered H_1 regime-dependent quantum advantage hypothesis is FALSIFIED across all eight non-fragile verdicts (Table VI). The LTC decomposition (Section IV B) localizes the mechanism in the quantum circuit; the H3 cross-tabulation (Section VI) finds a candidate expressivity-driven trend, underpowered at $n = 9$. Together these constitute a rigorous mechanistic null on a falsifiable quantum-ML hypothesis — the kind of result the Bowles–Schuld program calls for [16].

A. Limitations

Sample size. The forecaster task and the HPO-best sensitivity point run at $n = 9$; the combined ODE+PDE solver verdict at $n = 24$ (Section III C) is the largest. A larger ladder would tighten the CIs but, given the consistency across eight verdicts, is unlikely to shift the qualitative outcome.

Deferred systems. Two pre-registered systems were gate-tested but deferred for compute (amendment A11):

- **Kuramoto (12-dim)** – the per-component scalar circuit scales linearly with state dimension; a 12D system would cost $\approx$ 7 hr per cell. A single-cell smoke test confirmed the dispatch works.

- **KdV** ($u_t + 6uu_x + u_{xxx} = 0$) – the `jacrev`^{3} triple-nested autodiff through the QNode gate-tested (`results/p7_8_kdv_gate/gate_result.json`) *passes*: finite u_{xxx} at JIT compile time ≈ 4.6 s and per-point cost $\approx 0.43\times$ the second-order baseline via XLA fusion. Integrated training cost remains prohibitive at ≈ 12 s per gradient step.

Matched-capacity bound. Forecaster comparisons hold parameter count within a factor of two (Section II D). The cPINN solver baseline violates this by 3.3× (amendment A4) in the classical baseline's favor, and is therefore conservative for the FALSIFIED outcome.

Hyperparameter optimization. P7.6 ran symmetric HPO on the solver task at three anchor cells per family (LV s2, Van der Pol s1, Lorenz s2), three learning rates × two step budgets; its HPO-best $n = 9$ verdict (row 4 of Table VI) is the methodologically strongest sensitivity in the paper. The forecaster task uses default-Adam; an HPO-symmetric forecaster sensitivity is a natural revision-stage extension.

Simulator only. All quantum-circuit evaluations use `default.qubit` in PennyLane, which the pre-registration [4] explicitly disclaims hardware execution for. A small-scale hardware run (e.g., data-reuploading at $n_q = 4$) would be a natural revision-stage validation but does not address the falsified hypothesis.

Single ansatz per family. Each ansatz uses one layered configuration $(n_q, L) = (4, 1)$ at the default budget. Architectural variants (deeper reuploading at $L = 5$, alternative entanglement patterns) are deferred to the follow-up paper.

Multiple-comparison family. The eleven verdicts of Table VI form a single inferential family that the per-row $\alpha = 0.05$ CI machinery does not control. Under strict Holm–Bonferroni ($\alpha_{\text{family}} = 0.05$, $m = 11$), both PRIMARY headline verdicts retain FALSIFIED designation and the eight ancillary verdicts already crossed zero, so the correction is conservative; it is disclosed for transparency (Section II F and the Table VI footnote).

Latent-bug remediation. The post-peer-review audit (Amendments A20–A22) surfaced three issues that left the committed headline numbers intact: the `te_qpinn_fnn` readout-convention divergence (A20, requiring cell re-runs), the `brickwall` connectivity reduction at $n_q = 3$ (A21, disclosure-only strengthening of A18), and a 2D PDE hard-IC docstring omission (A22, latent; implementation always correct). The post-A20 re-runs are bundled into the Phase C Anvil sweep (Appendix A).

B. Future work: hardening quantum PINN training

The mechanism analysis (Section VI) identifies three interventions from the quantum-ML literature as plausibly relevant to the regime-contrast inversion:

1. **Quantum Natural Gradient** [21]. Adam treats the variational parameter manifold as Euclidean; QNG preconditions with the Quantum Fisher Information matrix, improving convergence and loss floors. The H3 KL-to-Haar trend $\rho = +0.518$ (Section VI B) is consistent with the expressivity-helps reading QNG targets.
2. **Causal training weighting** [35]. The PINN residual loss weights all collocation points equally, ignoring temporal ordering; an inverse-time-order schedule that grows weights only after upstream residuals decay improves training on stiff dynamics, directly the Van der Pol failure mode (Section IV A).
3. **Higher data-reuploading depth** [15]. H3 found Fourier $K_{\max} = 3$ at $L = 1$ across all ansätze — the ceiling at the current configuration. The Schuld–Sweke–Meyer relation $K_{\max} = L \cdot n$ predicts $K_{\max} = 15$ at $L = 5$. If the smooth-bin inductive bias the LTC decomposition isolated ($\Delta_{\text{liquid}} = +0.12$, Section IV B) scales with $K_{\max}$, deeper reuploading should recover the H_1 direction.

The negative finding and its mechanistic substructure (substrate-dependent τ signature, KL-to-Haar trend) seed three follow-up papers, listed by experimental readiness:

1. **The training-hardening paper.** The three interventions above (QNG [21], causal weighting [35], higher-$K_{\max}$ [15]) are tested jointly on the same hardness ladder — plus the deferred Kuramoto and KdV systems — at P7.6's HPO compute budget, with the same pre-registration and verdict machinery (locked in `.planning/P7_7_HARDENING_PREREG.md`). If any subset recovers H_1 in the predicted direction the paper reports it; otherwise it strengthens the present FALSIFIED verdict from one to four sensitivity points. Scope: ~ 60 cells, ~ 18 CPU-hours on Anvil.
2. **The τ-substrate mechanism paper.** The substrate-dependent τ signature (Figure 18; $\Delta_{\text{liquid}|\,\text{classical}} \approx +0.12$ vs. $\Delta_{\text{liquid}|\,\text{quantum}} \approx -0.33$, $|\text{gap}| = 0.45$) is the strongest mechanism finding here and is explained by neither substrate alone. The follow-up isolates the LTC τ machinery across substrate widths and reuploading depths, seeking a closed-form prediction for the sign of Δ_τ in the spirit of Hasani et al. [28]. Scope: ~ 36 cells.
3. **The domain-breadth paper (ChemE / BioE / PSE).** Extends the ladder from 8 dynamical-system archetypes to 12 process-engineering ODEs / PDEs spanning chemical-reactor (Van de Vusse, Saponification), biochemical (Monod, Yeast / Andrews, Contois), and process-systems (Fisher–KPP, CDR low/high-Pe, fixed-bed, ADM1) regimes, testing whether the FALSIFIED H1 generalizes to the domain-specific ladder a practitioner cares about. Scope: ~ 12 systems × 4 families × 3 seeds = 144 cells, ~ 36 CPU-hours.

All three inherit this paper's reproducibility infrastructure: pre-registration template, matched-capacity / paired-bootstrap / underfit-skyline-guard verdict machinery, and the `verify_paper_integrity` mechanical gate.

C. Positioning relative to the field

(1) The FALSIFIED outcome is not a uniqueness claim. The specific regime-dependent advantage formulated by Schuld–Killoran [4] does not materialize on this ladder at matched-capacity training. This neither shows that all quantum advantages on PINN-style tasks fail nor that the Schuld–Fourier hypothesis is wrong in general — the liquid-isolated decomposition ($\Delta = +0.12$) is the one verdict whose point estimate matches the original direction, which the higher-$K_{\max}$ intervention will probe.

(2) The mechanism attribution is the scientific content. A FALSIFIED pre-registered hypothesis is publishable [16], but its weight depends on what the falsification *shows*. Sections IV B and VI attribute the regime-contrast inversion to the quantum circuit — not the liquid-τ machinery, not baseline-tuning artifacts — a contribution to understanding when matched-capacity quantum-PINN training fails.

(3) The positive sub-findings are independent of the aggregate verdict. `te_qpinn_qnn` achieves a 24×-tighter seed standard deviation and 2.5× lower mean relative-L^2 than the matched classical PINN on FitzHugh–Nagumo (Section III D); `rf_qrc` achieves the strongest cell on Lorenz s0 and s2 (Section IV A). Both occur at zero classical parameters in the quantum-cell stack — regime-specific structural advantages the falsification does not contradict, of the kind the `te_qpinn_qnn` [11] and `rf_qrc` [14] original works target, here seen in an independent benchmark.

(4) Comparison to classical operator-learning competitors. Our matched-classical baselines are MLP-PINNs, mirroring the PINN-vs-PINN framing of the quantum-PINN literature itself. Benchmarking against operator-learning state-of-the-art — DeepONet [36] and the Fourier Neural Operator [37] — would collapse the PINN inductive-bias question (trial-solution structure) into a representational-capacity question (data scale and architecture); the follow-up paper is the appropriate venue for such baselines, paired with the corresponding generalization analysis [3].

D. Reproducibility and audit

Every reported number is gated mechanically by `scripts/verify_paper_integrity.py`, which matches committed JSON files against the paper's text; a reviewer running it after cloning the repository (https://github.com/shawngibford/qlnn) receives a pass/fail summary on every numerical claim, including the LTC decomposition's algebraic-identity check (Section IV B). Twenty-two amendments (A1–A22) are documented in `PRE_REG_AMENDMENT.md` with per-amendment empirical findings: A15–A19 (2026-05-28 internal audit) propose budget-parity, ansatz-aliasing, and sub-experiment refinements; A20–A22, filed the same day after a five-reviewer pass, record a paired QPINN readout divergence, a brick-wall connectivity diagnosis, and a latent docstring defect. The verdict refresh under the A15–A22 configuration is deferred to a supplementary compute campaign.

The full reproduction pipeline (`scripts/reproduce_paper.sh`) regenerates every cell from scratch on a single MacBook Pro M1 in $\approx$ 8 hours, parallelizable across sweep scripts; the 36 P7.11 non-liquid quantum cells reproduce in under 30 minutes.

VIII. CONCLUSIONS

We pre-registered and executed a quantum PINN benchmark testing the Schuld–Fourier regime-dependent advantage hypothesis [4, 15]: four quantum-PINN families on a hardness ladder of four ODEs and four PDEs at three seeds each, against five matched classical baselines and the complete 2×2 fairness matrix on the forecaster task.

The pre-registered H_1 regime-dependent quantum advantage is FALSIFIED. Across all eight non-fragile sensitivity points, the paired-bootstrap 95% confidence intervals on $\Delta_{\text{smooth}} - \Delta_{\text{broad}}$ either include zero or exclude it negatively. The primary solver verdict at $n = 24$ has $\Delta_{\text{diff}} = -0.084$, CI $[-0.278, +0.061]$; the primary forecaster verdict at $n = 9$ has $\Delta_{\text{diff}} = -0.501$, CI $[-0.804, -0.244]$, excluding zero in the direction opposite the prediction.

The LTC decomposition attributes the inversion to the quantum circuit. The quantum-isolated $\Delta_{\text{diff}} = -0.616$ (CI $[-1.167, -0.178]$) is significantly negative, while the liquid-isolated $\Delta_{\text{diff}} = +0.115$ (CI $[-0.097, +0.377]$) is the only verdict whose point estimate matches the Schuld–Fourier direction: the classical liquid-time-constant machinery carries the predicted smooth-bin inductive bias, and the quantum circuit destroys it.

One positive sub-finding survives. At zero classical parameters in the quantum cell, `te_qpinn_qnn` on FitzHugh–Nagumo attains 24× tighter seed standard deviation and 2.5× lower mean error than the matched cPINN. The H3 cross-tabulation flags expressivity (KL-to-Haar distance) as a candidate within-roster predictor of per-cell Δ ($\rho = +0.518$, LOO-robust at $n = 9$), underpowered but directionally consistent with the decomposition's mechanism attribution.

The methodological contribution is the framework itself. The open-source pipeline at https://github.com/shawngibford/qlnn implements the Bowles–Schuld [16] program end to end, and its P7.11 ablation completes the forecaster fairness 2×2 – to our knowledge the first quantum-PINN benchmark to do so.

Future work targets the mechanism. The follow-up paper applies three interventions the mechanism analysis identifies as relevant: Quantum Natural Gradient optimization [21] for the quantum-circuit training inefficiency, causal training weighting [35] for the stiff-dynamics failure mode, and data-reuploading depth $L = 5$ for the Fourier

$K_{\max}$ ceiling (Section VI A); the deferred Kuramoto and KdV systems are queued alongside. Whether any recovers the H_1 direction is an empirical question the framework introduced here is built to keep falsifiable.

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